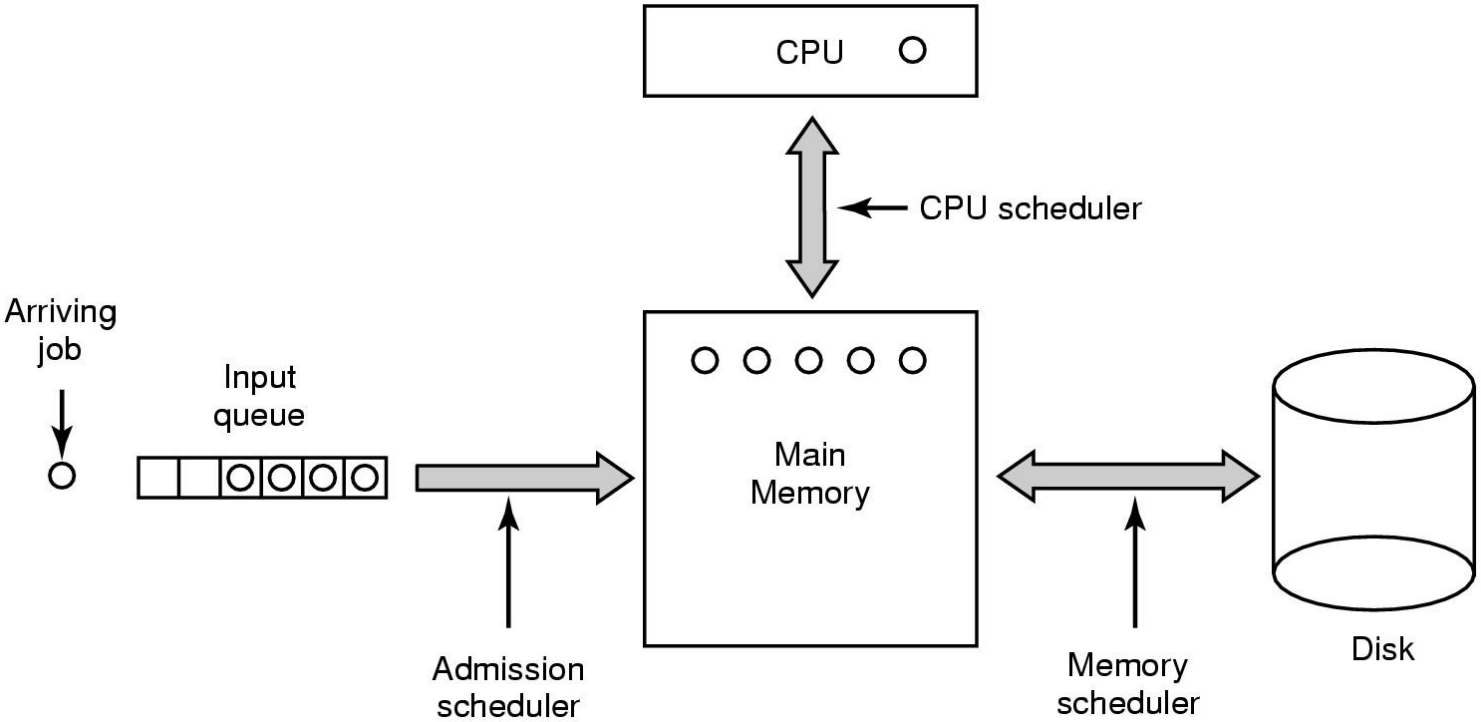
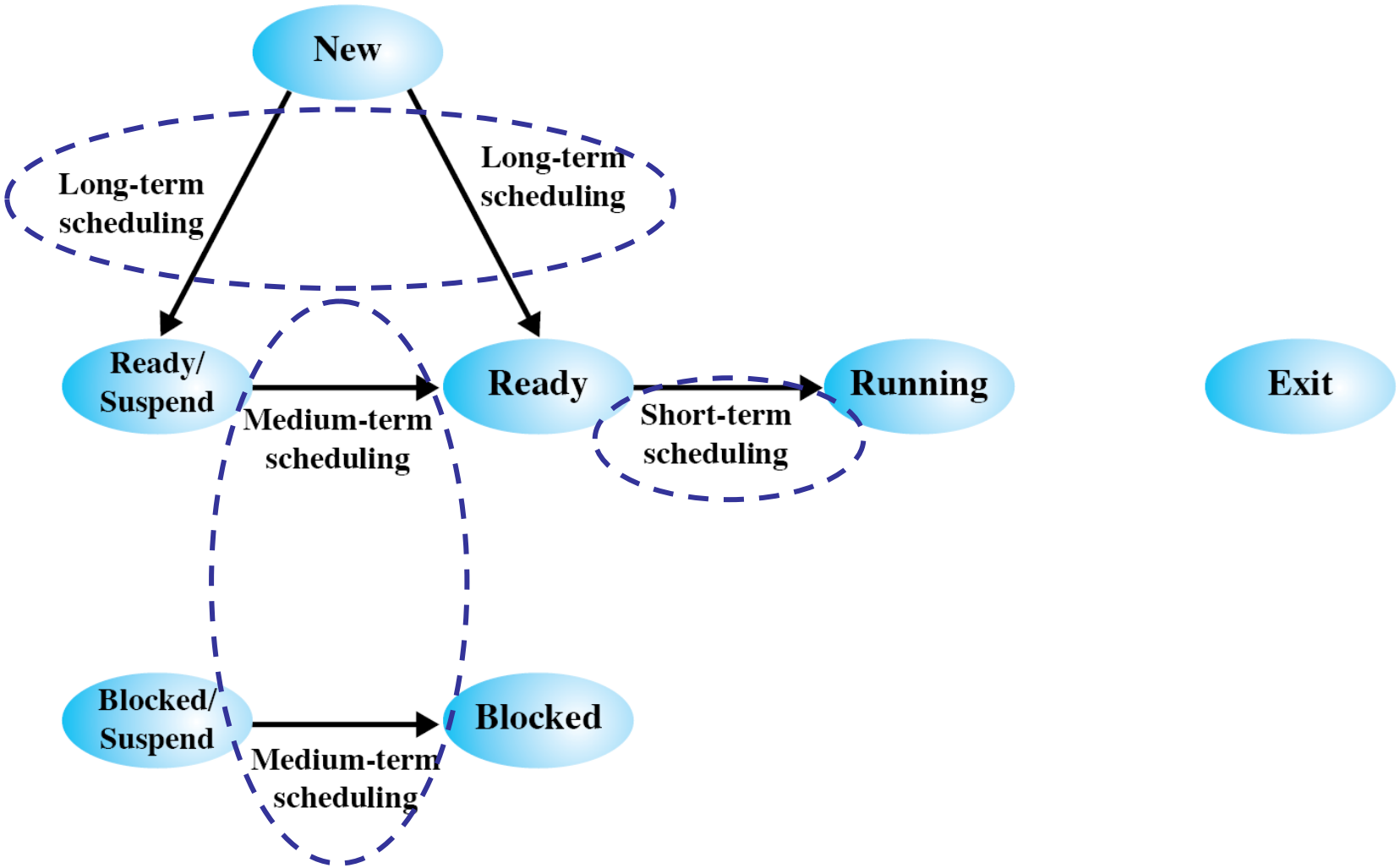
Operating Systems

Process and Thread (KLT) Scheduling

Three Level Scheduling

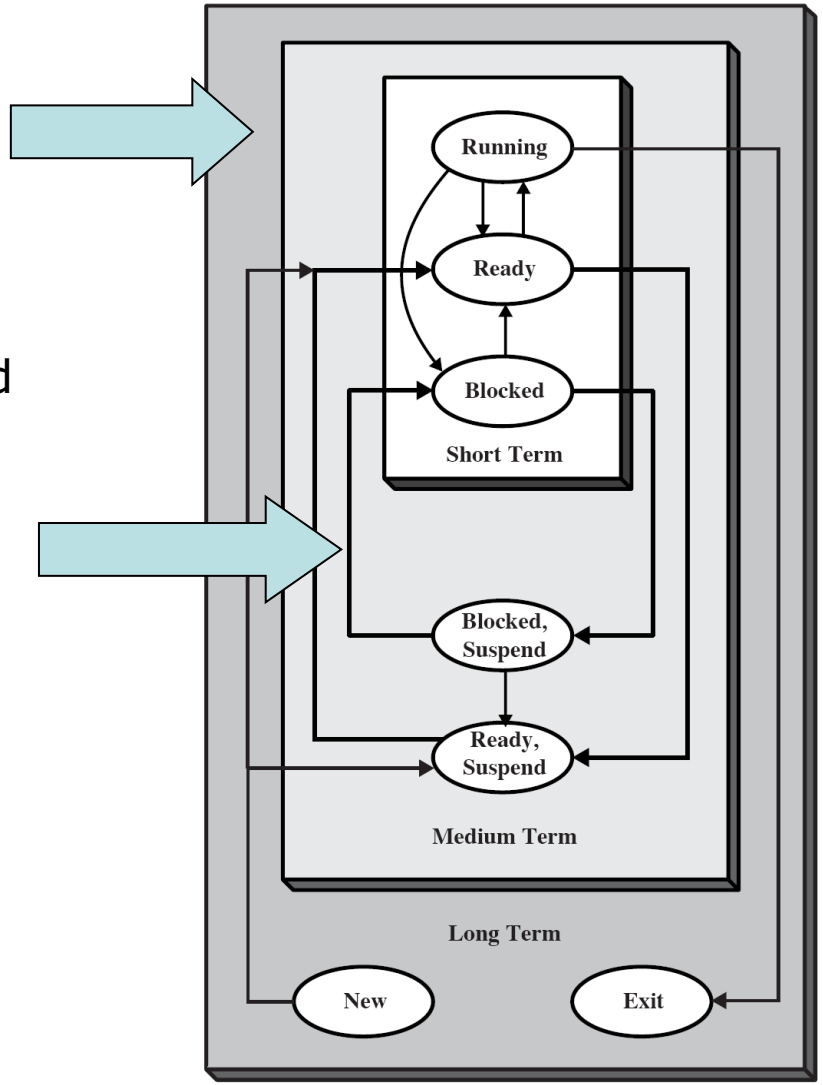


Scheduling and Process Transitions



Long- and Medium-Term Schedulers

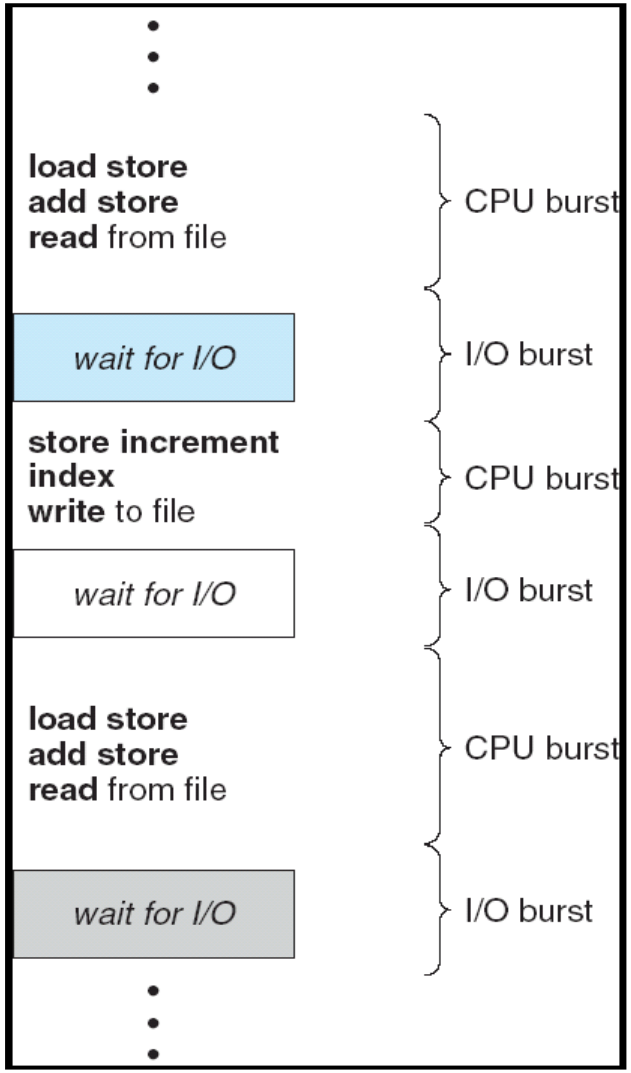
- Long-term scheduler
  - Determines which programs are admitted to the system
    - I.e., to become processes
  - Requests can be denied
    - E.g., in case of thrashing or overload
- Medium-term scheduler
  - Decides when/which processes to suspend/resume
- Both control the degree of multiprogramming
  - More processes, smaller percentage of time each process is executed



CPU–I/O Burst Cycle

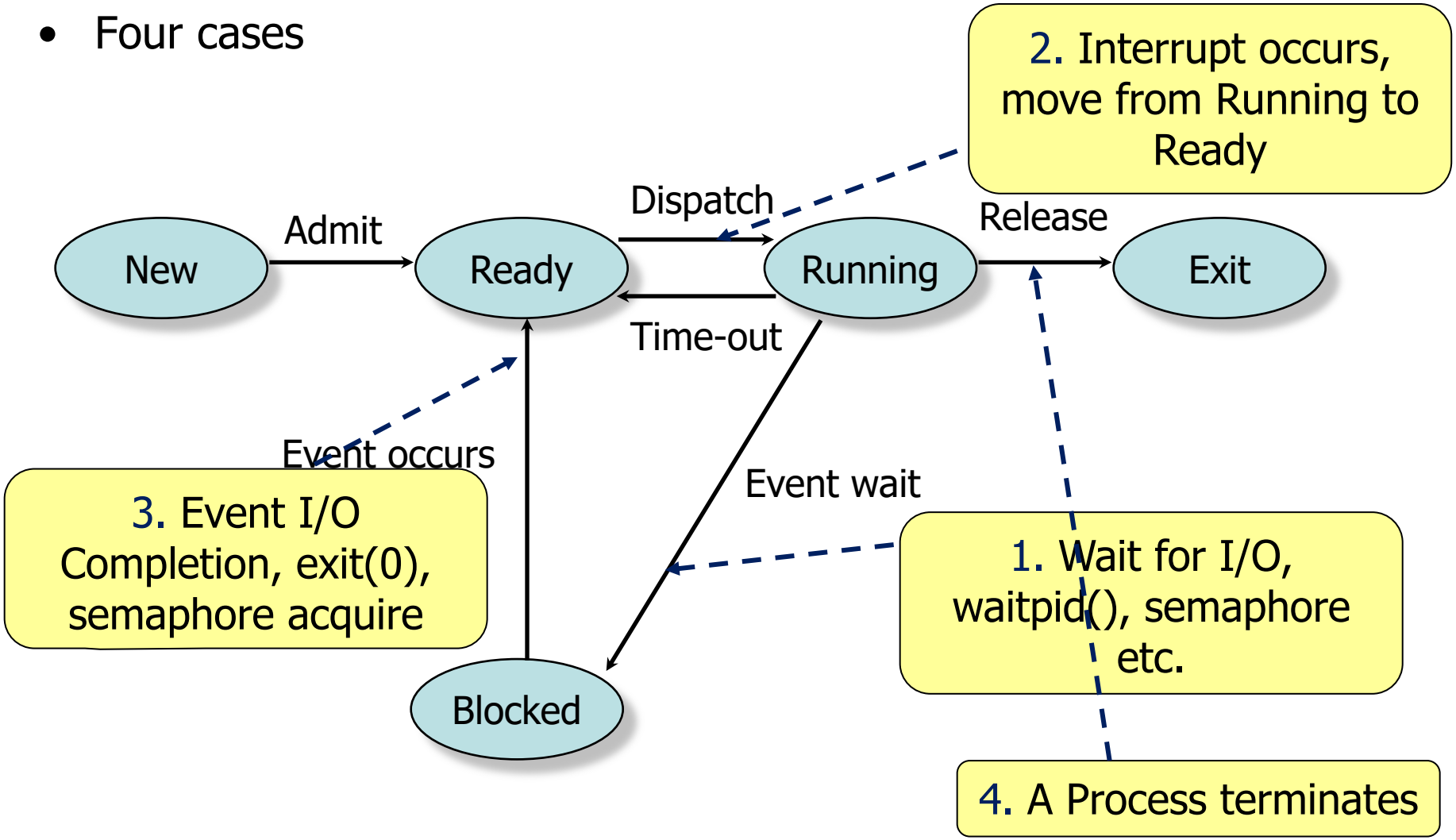
- Process execution consists of a cycle of
  - CPU execution and
  - I/O wait

- A process may be
  - CPU-bound : Long CPU bursts
  - IO-bound : Short CPU bursts



Short-Term Scheduler: Selection of a New Process

- Four cases



Decision Mode

Non preemptive

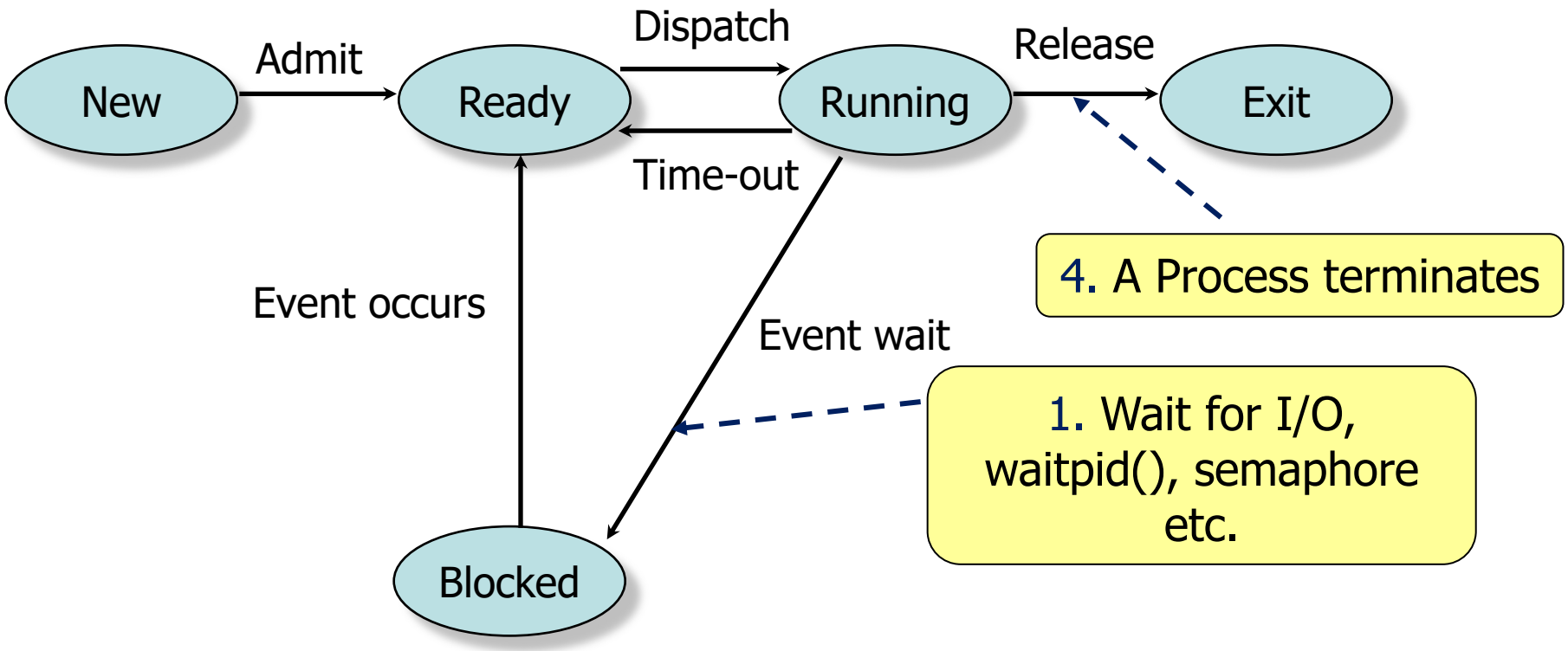
- Once a process is in the running state, it will continue until it terminates or blocks itself for I/O, child process, semaphore etc.

Preemptive

- Currently running process may be interrupted
- Now transition from Running to Ready state possible
- Prevents a single process to monopolize the processor for very long

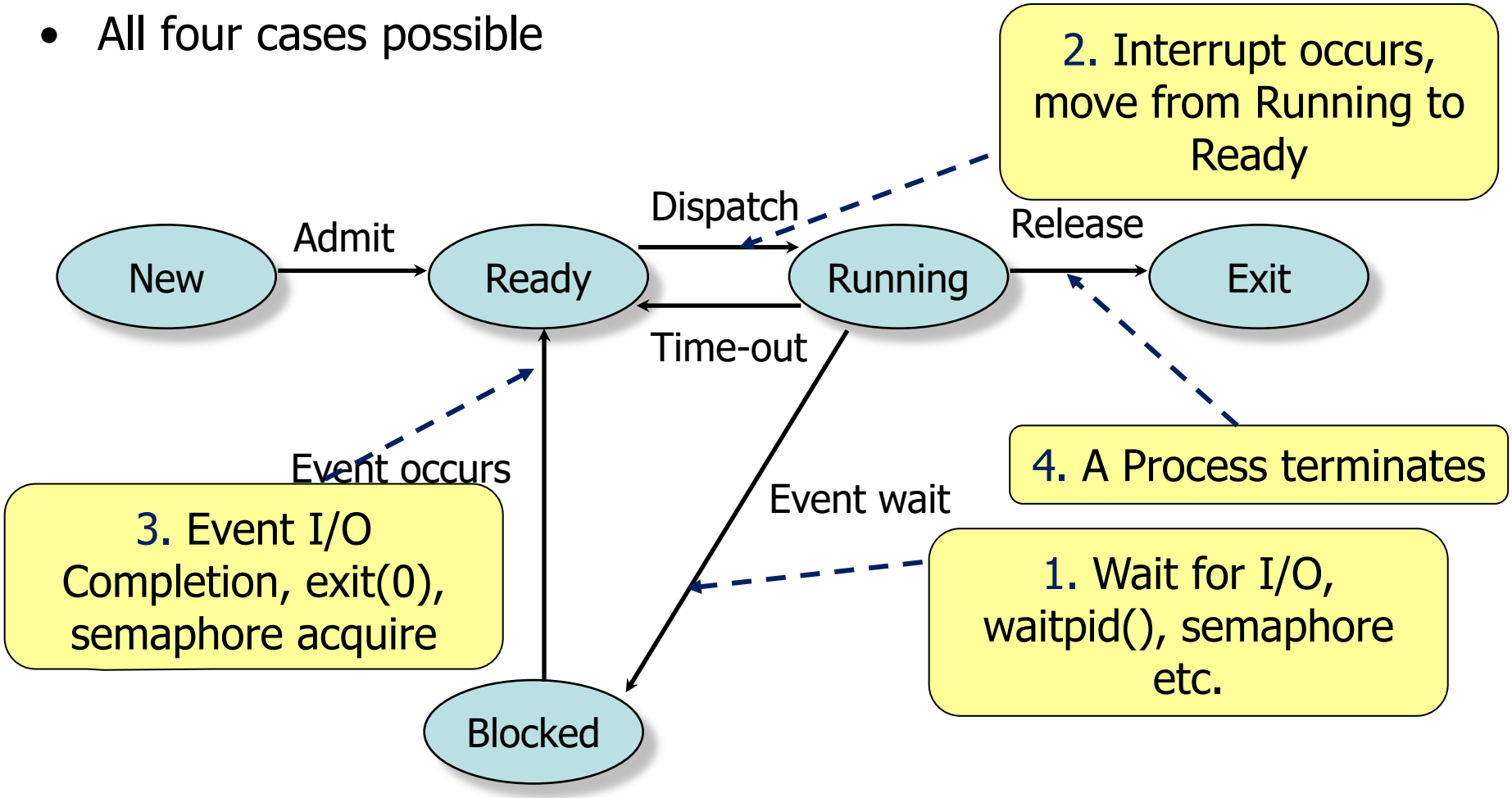
Non Preemptive Scheduling

- Only the case 1 and 4



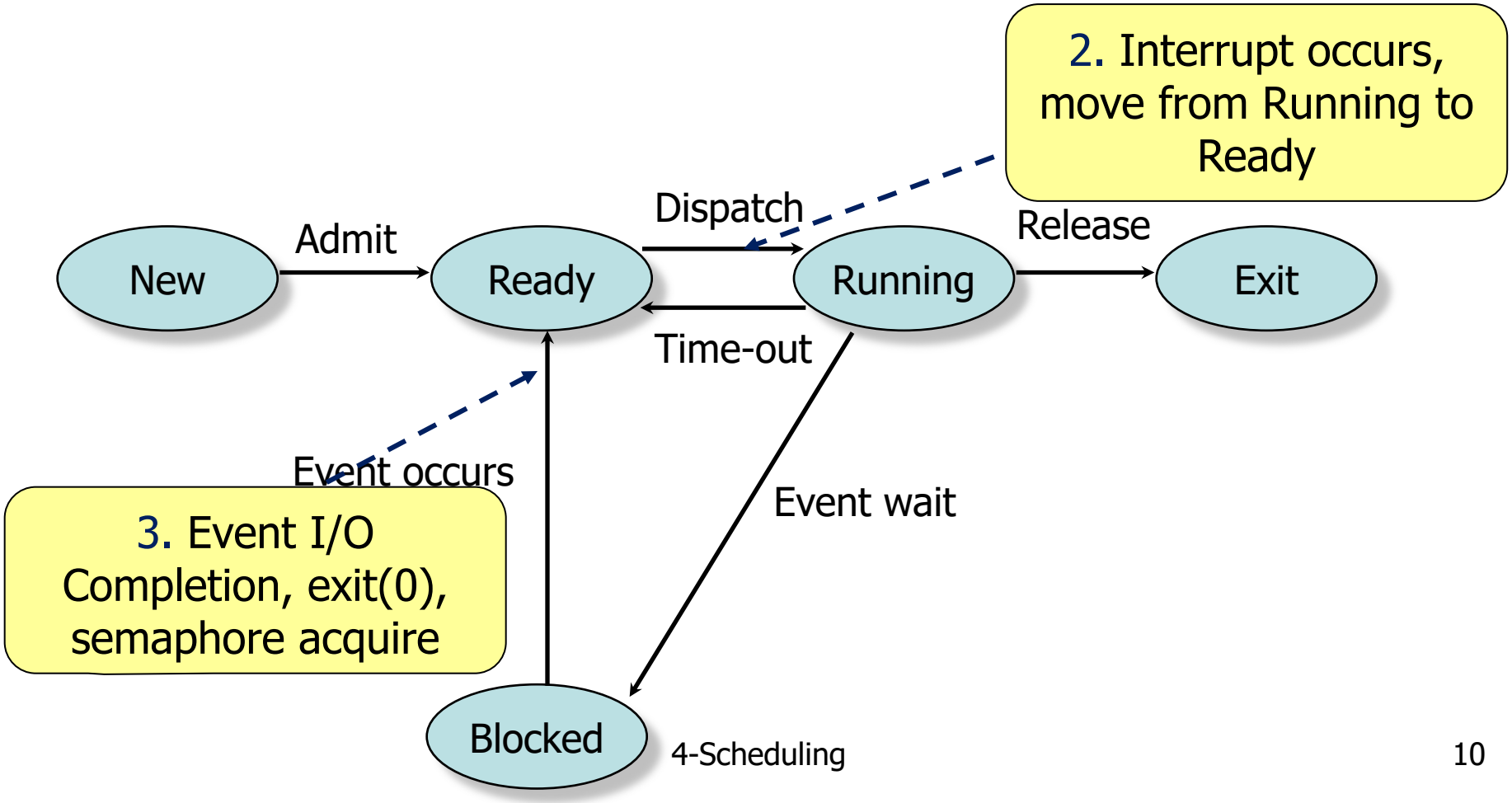
Preemptive Scheduling

- All four cases possible



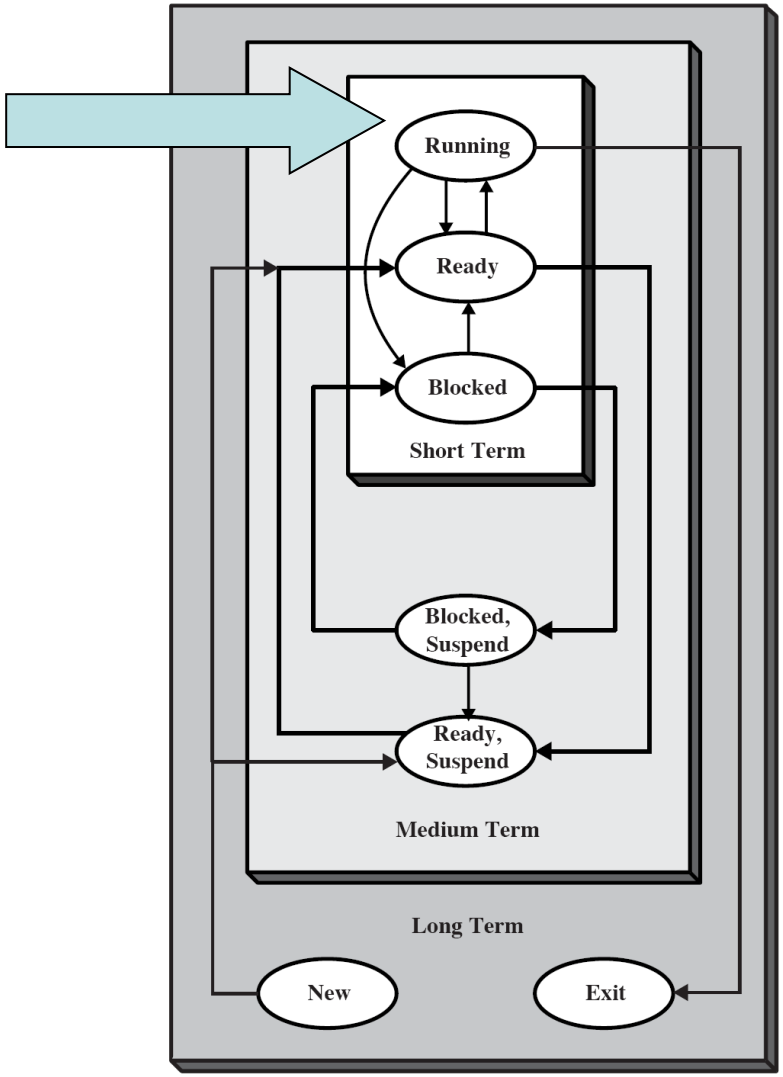
Preemptive Scheduling

- In case of 2 and 3, there is a choice
  - Whether to continue, with the same process
  - Or select a new one from the ready queue



Short-Term Scheduler

- Decides which process will be dispatched
  - Invoked upon Clock interrupts, I/O interrupts, Operating system calls, Signals
- Dispatch latency: Period of time the dispatcher needs to stop one process and start another running
- Dominating factors
  - Switching context
  - Selecting a process to dispatch
- Arrival time: Time when a process is admitted to the system
- Service Time: Period of time a process executes in running state



Scheduling Issues

- Fairness
  - Don’t starve process
- Priorities
  - Most important first
- Deadlines
  - Task X must be done by time t
- Optimization
  - Throughput, response time
- Scheduler Efficiency
  - Overhead, e.g., context switching, computing priorities, …
- Reality - No universal scheduling policy
  - Many models

Scheduling Criteria and Optimization goals

- CPU utilization
  - Percentage of time CPU is busy
  - Keep CPU as busy as possible

- Throughput
  - No. of processes that complete their execution per time unit

- Turnaround time (TAT)
  - Amount of time to execute a particular process
  - TAT = Time of completion – Arrival time
    - Execution + all the waiting for resources including CPU
    - Involves also the IO schedulers
  - Not a good criteria for interactive systems

Scheduling Criteria and Optimization goals

- Waiting time
  - Waiting Time = Sum of the periods spent waiting in the Ready queue
  - Scheduling Algorithm does not effect the waiting time in Block queue
    - Only effect the waiting time in the Ready queue

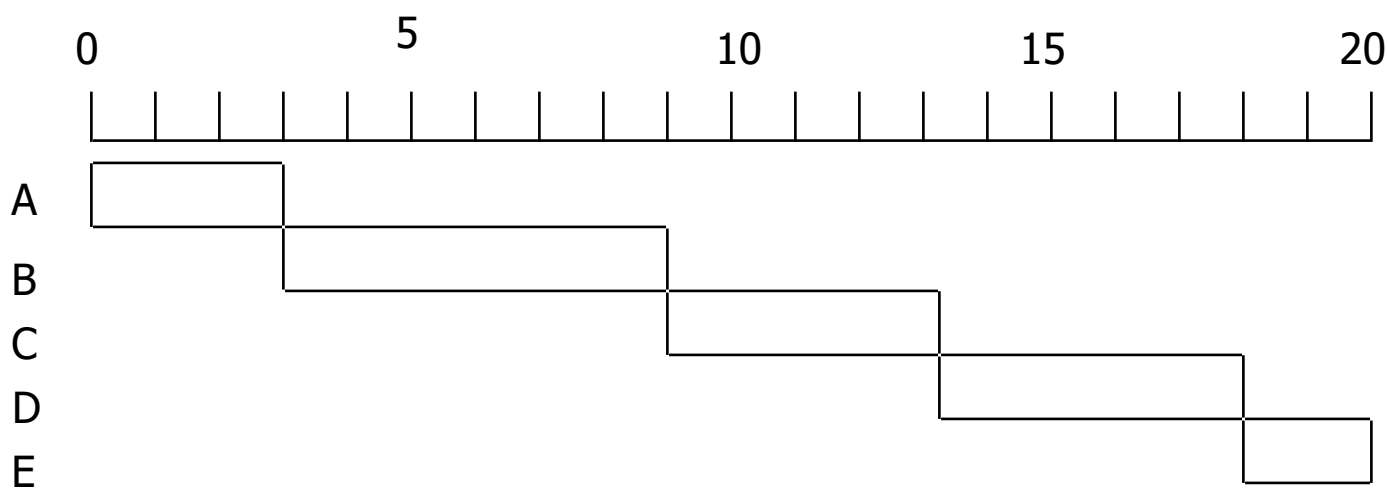
- Response time
  - Amount of time from submission of the request until first response
  - Response time = First response – Arrival time
    - Execution + waiting time in ready queue

- Summary
  - Maximize: CPU utilization, Throughput
  - Minimize: Turnaround time, Waiting time, Response time

First-Come-First-Served (FCFS)

- Non-preemptive
- Favors CPU-bound processes
- A short process may have to wait very long before it can execute
- Convoy effect – Many I/O processes may stuck behind CPU-bound process

Process	Arrival Time	Burst Time
A	0	3
B	2	6
C	4	4
D	6	5
E	8	2



First-Come-First-Served (FCFS)

- Wait time depends on arrival order
- Worst case: long job arrives first
- Example: Three processes with service times: A=100, B=1, C=2
  - Case 1: Processes arrive in the order A, B, C

Average Waiting Time = (0 + 100 + 101) / 3 = 67

  - Case 2: Processes arrive in the order B, C, A

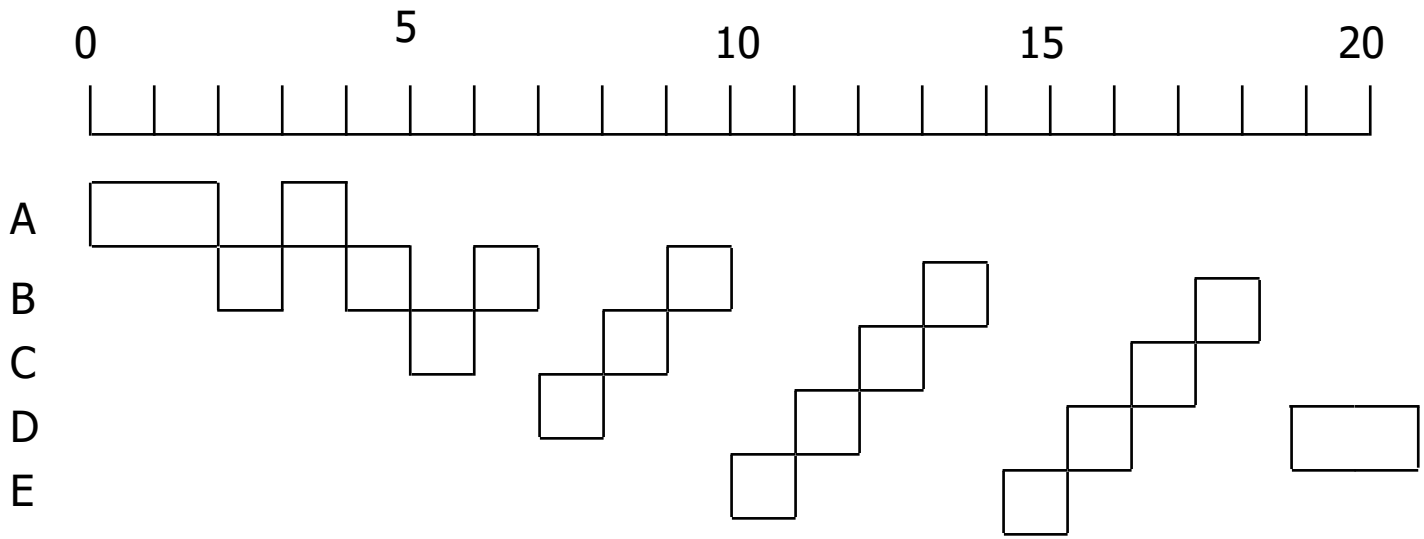
Average Waiting Time = (0 + 1 + 3) / 3 = 1.33

First-Come-First-Served (FCFS) – Convoy Effect

Consider one CPU-bound and many I/O-bound processes

1. The CPU intensive process blocks the CPU
2. A number of I/O intensive processes stuck behind this process
   - Leaving the I/O devices idle
   - Low I/O device utilization
3. When the CPU-bound process finally issues I/O request
   - I/O processes pass through the CPU quickly
   - Leaving the CPU idle while everyone queues up for I/O
4. The cycle repeats itself when the CPU intensive process gets back to the ready queue

Round Robin



Process	Arrival Time	Burst Time
A	0	3
B	2	6
C	4	4
D	6	5
E	8	2

- Preemption based on clock (interrupts on time slice or quantum -*q*- usually 10-100 msec)
- Fairness: for *n* processes, each gets `1/n` of the CPU time in chunks of at most `q` time units
- Performance
  - q large ⇒ FCFS
  - q small ⇒ overhead can be high due to context switches

Round Robin – Example with Time Quantum = 4

Process	Burst time
P_1	24
P_2	3
P_3	3

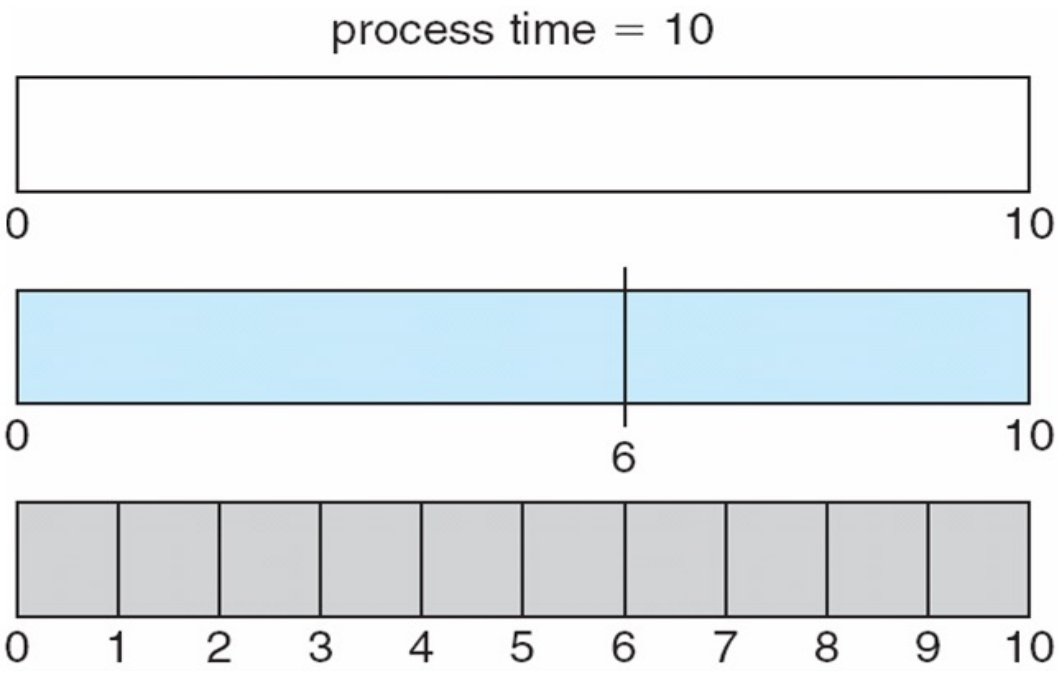
- Process may have CPU burst of less than 1 time quantum
  - Process itself release the CPU voluntarily
  - Dispatcher will proceed to the next process in ready queue

P_1	P_2	P_3	P_1	P_1	P_1	P_1	P_1

0 4 7 10 14 18 22 26 30

Time Quantum and Context Switch Time

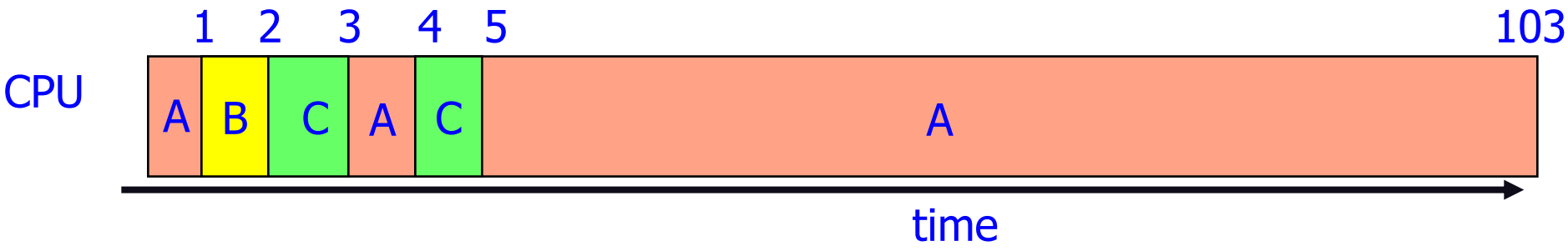
- Quantum –q- should be large compared to context switch time
- Quantum –q- usually 10ms to 100ms, context switch < 10 usec



quantum	context switches
12	0
6	1
1	9

Round Robin

- Low average waiting time when job lengths vary
- Example: Three processes with service times: A=100, B=1, C=2



- Average waiting time
  - (3 + 1 + 3) / 3
- Average completion (Turn around time)
  - (103 + 2 + 5) / 3

Round Robin – Disadvantage

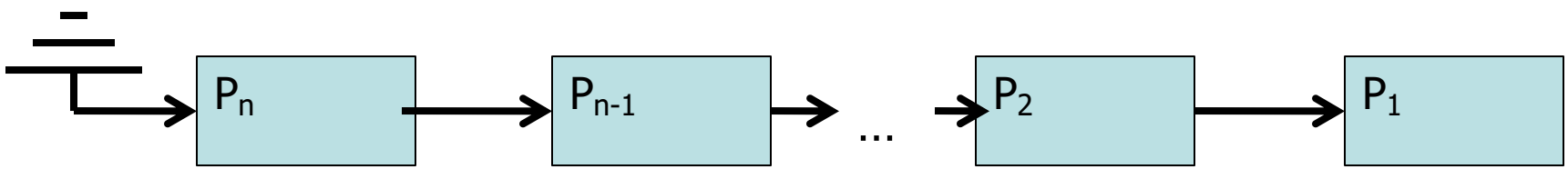
- Good for Varying sized jobs
- But what about same-sized processes?
- Example: Two processes with same service time =100

- Average completion time, i.e., Turn Around Time (TAT)?
  - (200 + 200) / 2 = 200
- How does this compare with FCFS for same two jobs?
  - (100 + 200) / 2 = 150

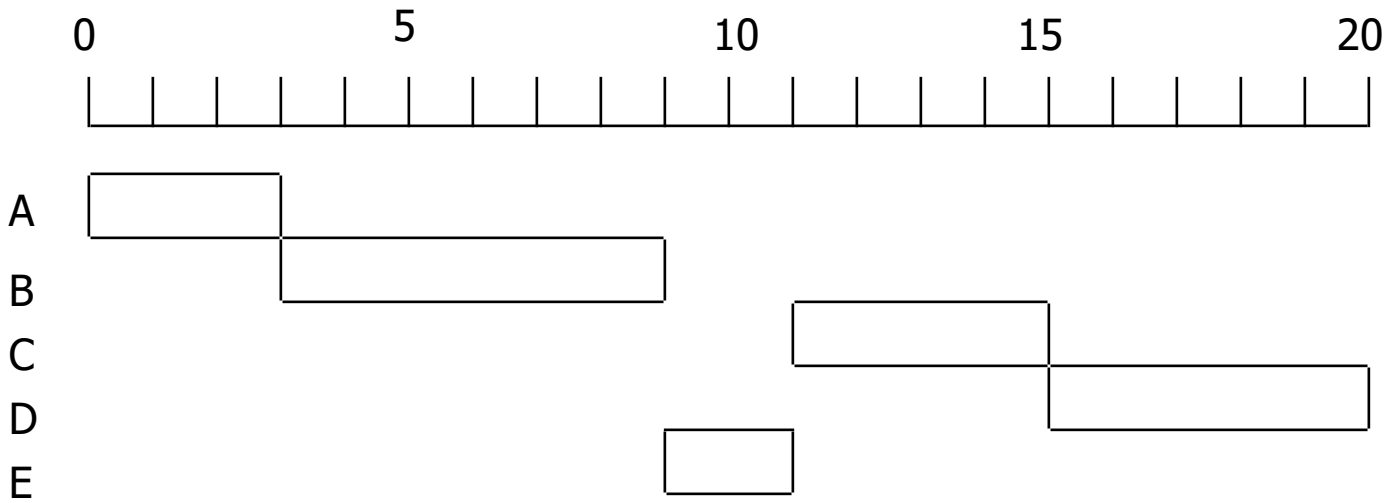
Dispatcher Performance: Round Robin and FCFS

- Maintain linked list
  - Dequeue: Remove head
  - Enqueue: Append tail

- Cost
  - Enqueue: O(1)
  - Dequeue: O(1)



Shortest Job First (SJF) – Non-preemptive

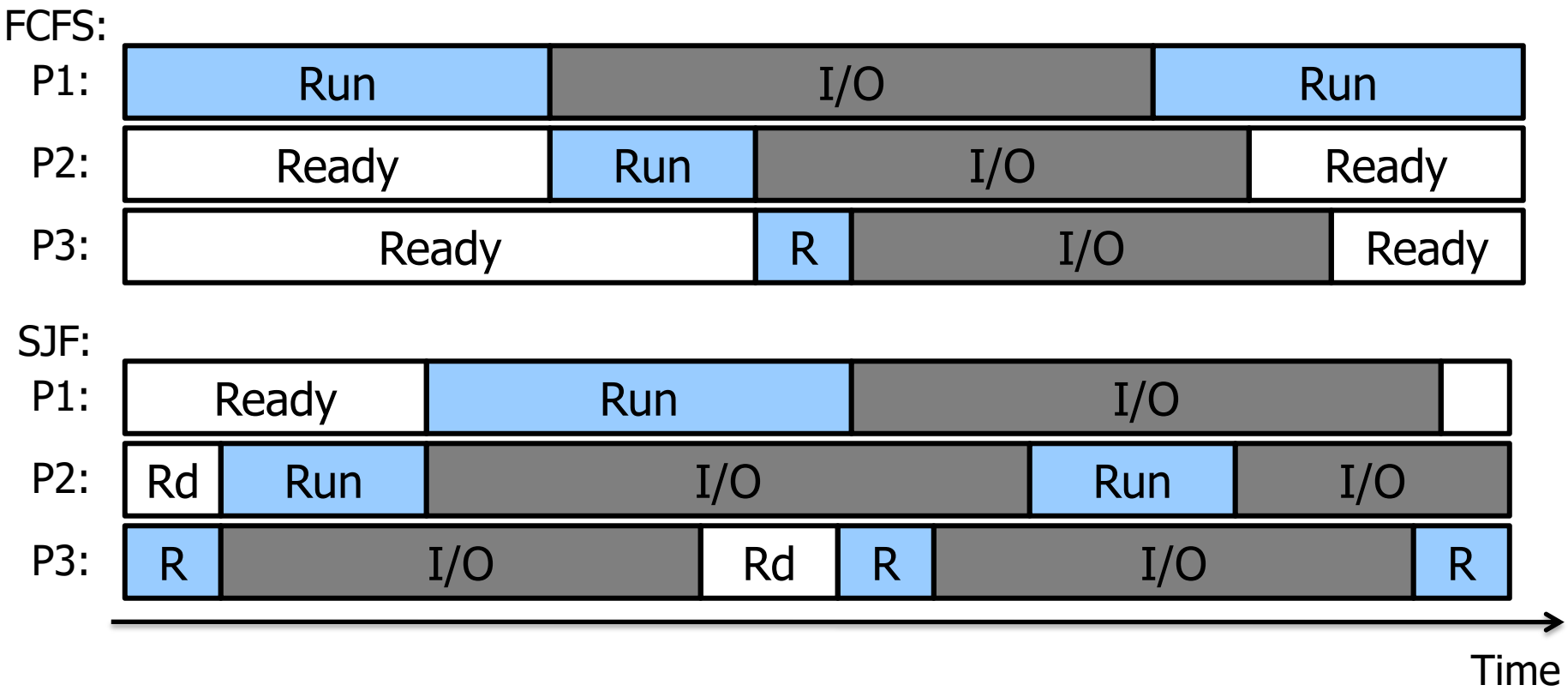


Process	Arrival Time	Burst Time
A	0	3
B	2	6
C	4	4
D	6	5
E	8	2

- Associate with each process the length of its next CPU burst
  - Short process jumps ahead of longer processes
- Avoid convoy effect

Performance of SJF Scheduling

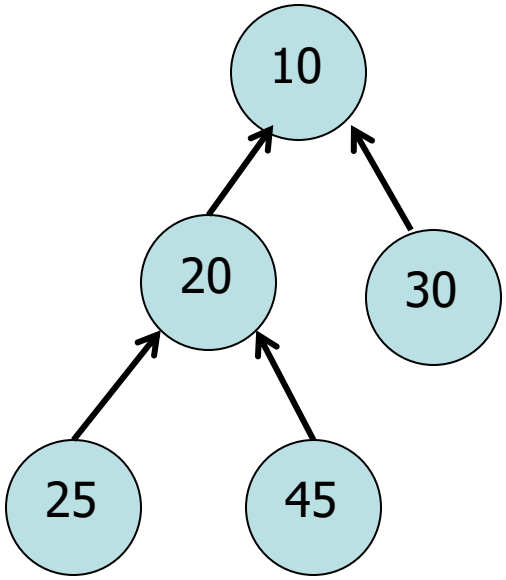
- Gives high throughput
- Gives minimum average waiting time for a given set of processes



Dispatcher Performance: SJF

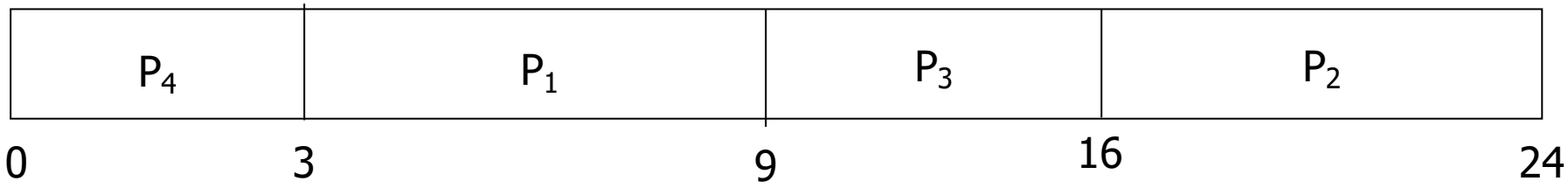
- Dispatcher utilizes heap operations
  - Enqueue: `insert_heap(`P_i`, `t_i` )`
  - Dequeue: `del_min()`
  - Decrease the priority dynamically

- Performance depends on implementation
  - Binary Heap
    - Insert : $O(\log_2 \quad n)$
    - Dequeue: $O(\log_2 \quad n)$



SJF (Non-preemptive) – Example

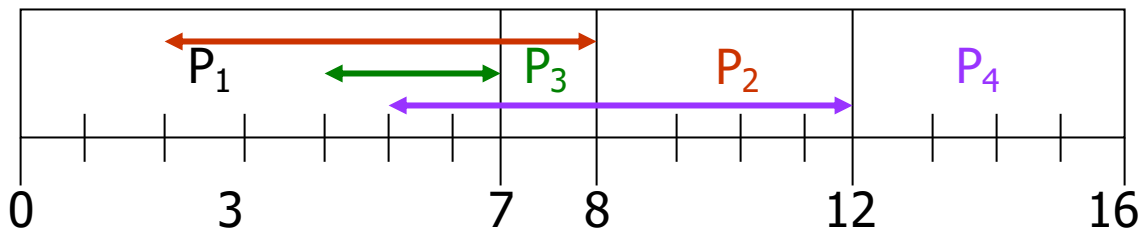
Process	Burst time
P_1	6
P_2	8
P_3	7
P_4	3



- Average waiting time = (3 + 16 + 9 + 0) / 4 = 7

SJF (Non-preemptive) – Example

Process	Arrival time	Burst time
P_1	0	7
P_2	2	4
P_3	4	1
P_4	5	4



- Average waiting time
  - (0 + 6 + 3 + 7)/4 = 4

Shortest Remaining Time First

- Shortest Job First (SJF) can be preemptive or non-preemptive
  - Preemptive SJF is also called Shortest Remaining Time First

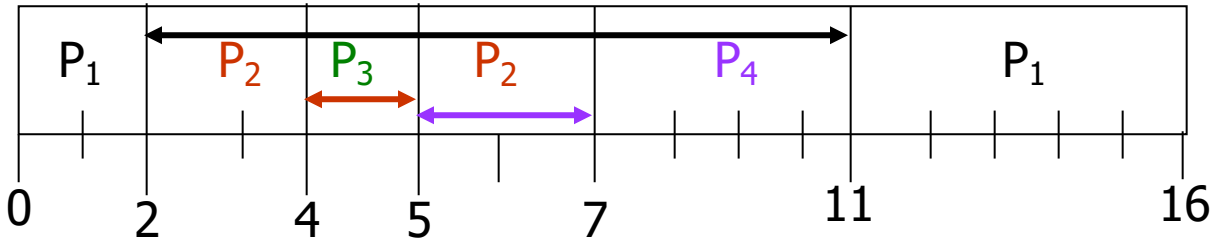
Overall Idea

- While an existing process is executing
  - A new process arrives at the ready queue

- Existing process is preemptive
  - If the CPU burst of the newly arrived process is shorter than the remaining execution time of existing process

Shortest Remaining Time First – Example

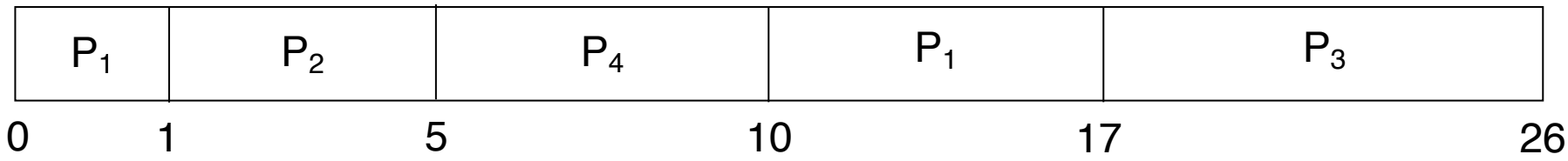
Process	Arrival time	Burst time
P_1	0	7
P_2	2	4
P_3	4	1
P_4	5	4



- Average waiting time
  - (9 + 1 + 0 + 2)/4 = 3

Shortest Remaining Time First – Example

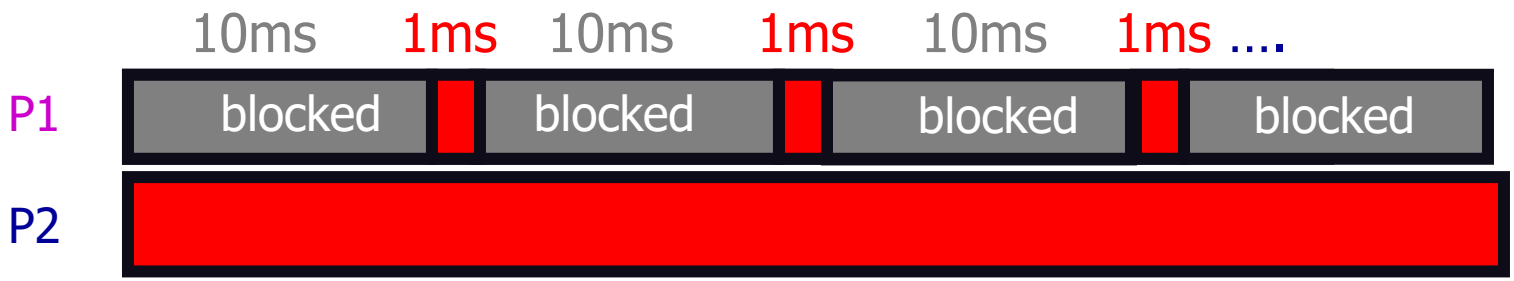
Process	Arrival time	Burst time
P_1	0	8
P_2	1	4
P_3	2	9
P_4	3	5

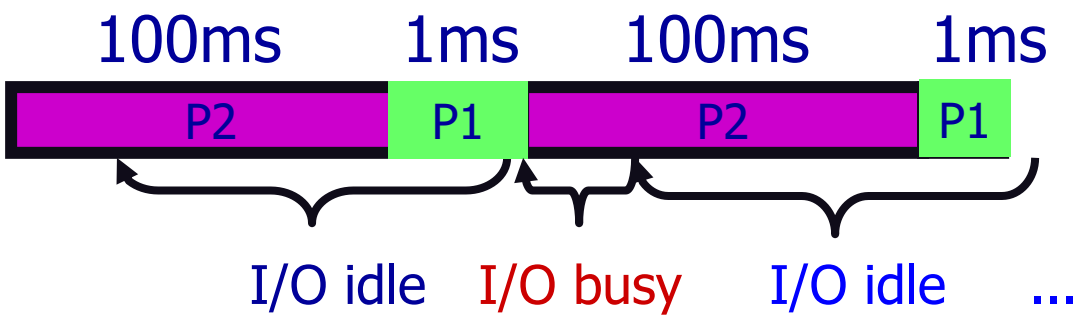


- Average waiting time
  - [(10-1)+(1-1)+(17-2)+5-3)]/4 = 26/4 = 6.5

SJF vs. RR

- Two processes P1, P2
  - Round robin with 100ms time slice





- SJF Offers better I/O utilization

Properties of SJF

- Possibility of starvation for longer processes
- Must estimate length of next CPU burst
  - Need for an algorithm that adapts the estimate dynamically
  - Algorithm must deal with variable length of CPU bursts
    - Exponential averaging
- What if estimates are not correct?
  - Preemption by the operating system

Priority Scheduling

Scheduler can choose a process of higher priority over one of lower priority

- Can be preemptive or non-preemptive
- Priorities can be static or dynamic
- E.g., in SJF priority is the predicted next CPU burst time

Problem of priority based schemes: Danger for starvation

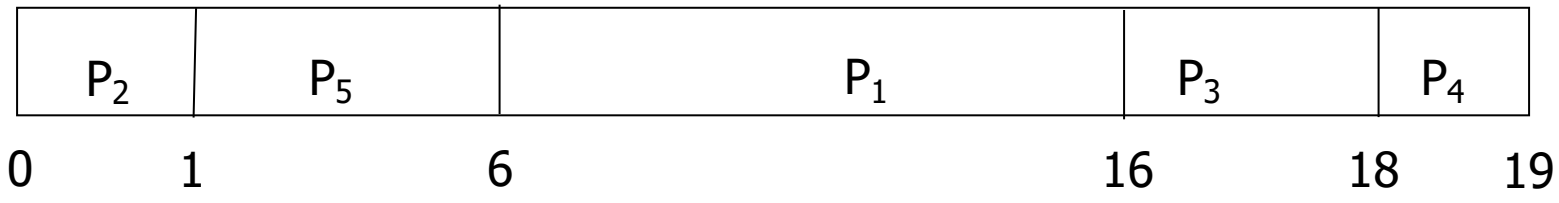
- Low priority processes may never execute
- E.g., process with a long CPU burst in SJF may never execute

A solution: Aging

- As time advances the priority of a process is increased
- Longest running processes eventually reaches highest priority

Priority Scheduling – Example

Process	Priority	Burst time
P_1	3	10
P_2	1	1
P_3	4	2
P_4	5	1
P_5	2	5



- Average waiting time = 8.2

Priority Scheduling

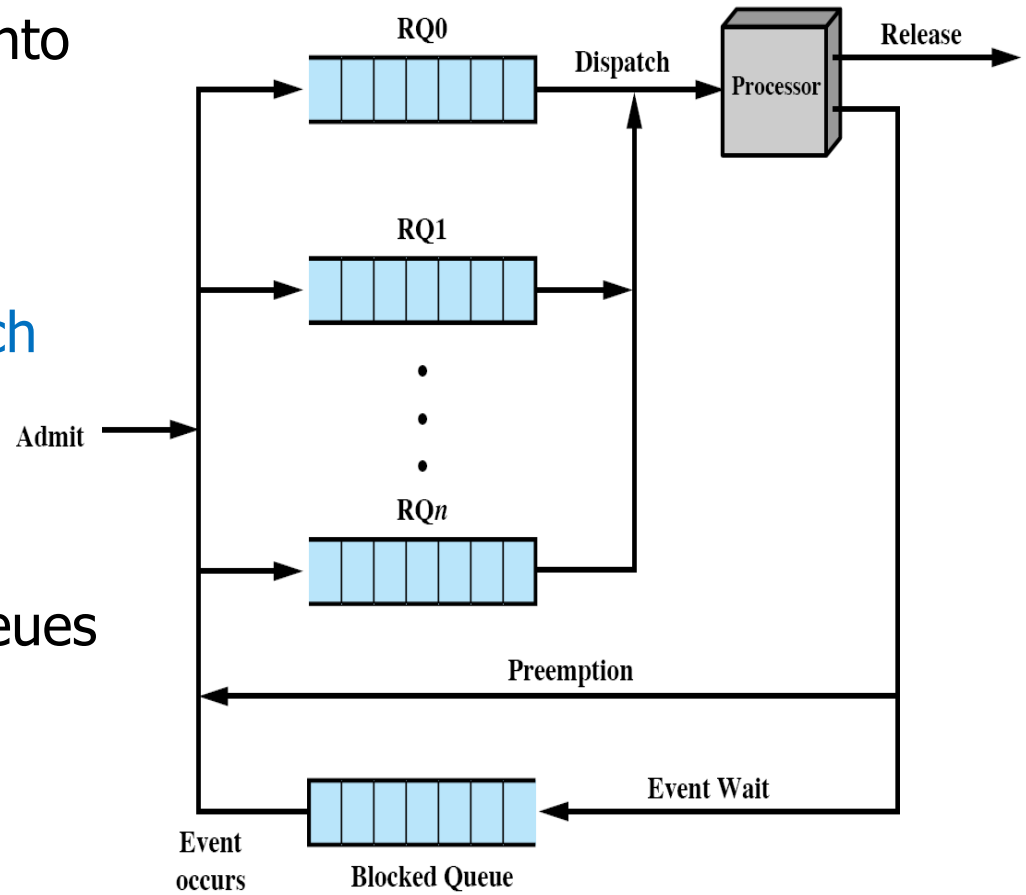
- If processes have equal priority
  - Round robin
  - FCFS

- I/O Bound vs. CPU Bound
  - Which type of processes should be given Higher Priority
  - In order to keep I/O busy increase priority for jobs that often block on I/O

Multilevel Queue

- Idea: Reduce dispatch latency by multilevel queues

- Ready queue is partitioned into separate queues, e.g.,
  - Foreground (interactive)
  - Background (batch)
- Scheduling algorithm for each queue, e.g.,
  - Foreground → RR
  - Background → FCFS
- And scheduling between queues …



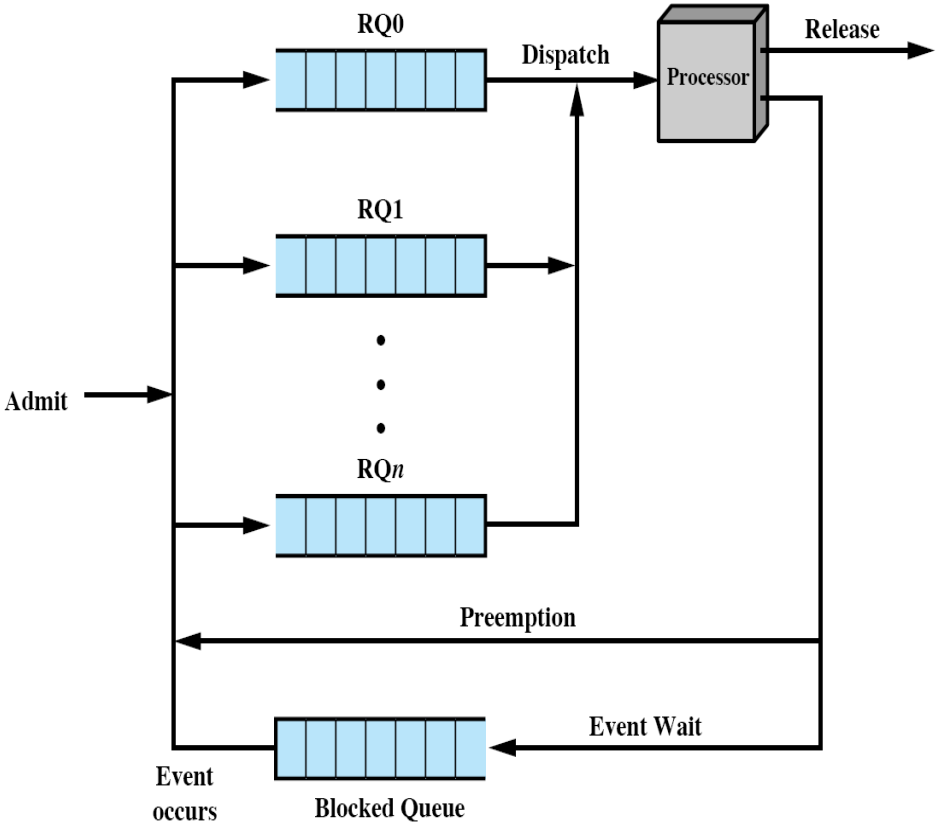
Multilevel Queues: Scheduling Between the Queues

Fixed order/priority e.g.,

- Serve all from foreground then from background
- Possible starvation

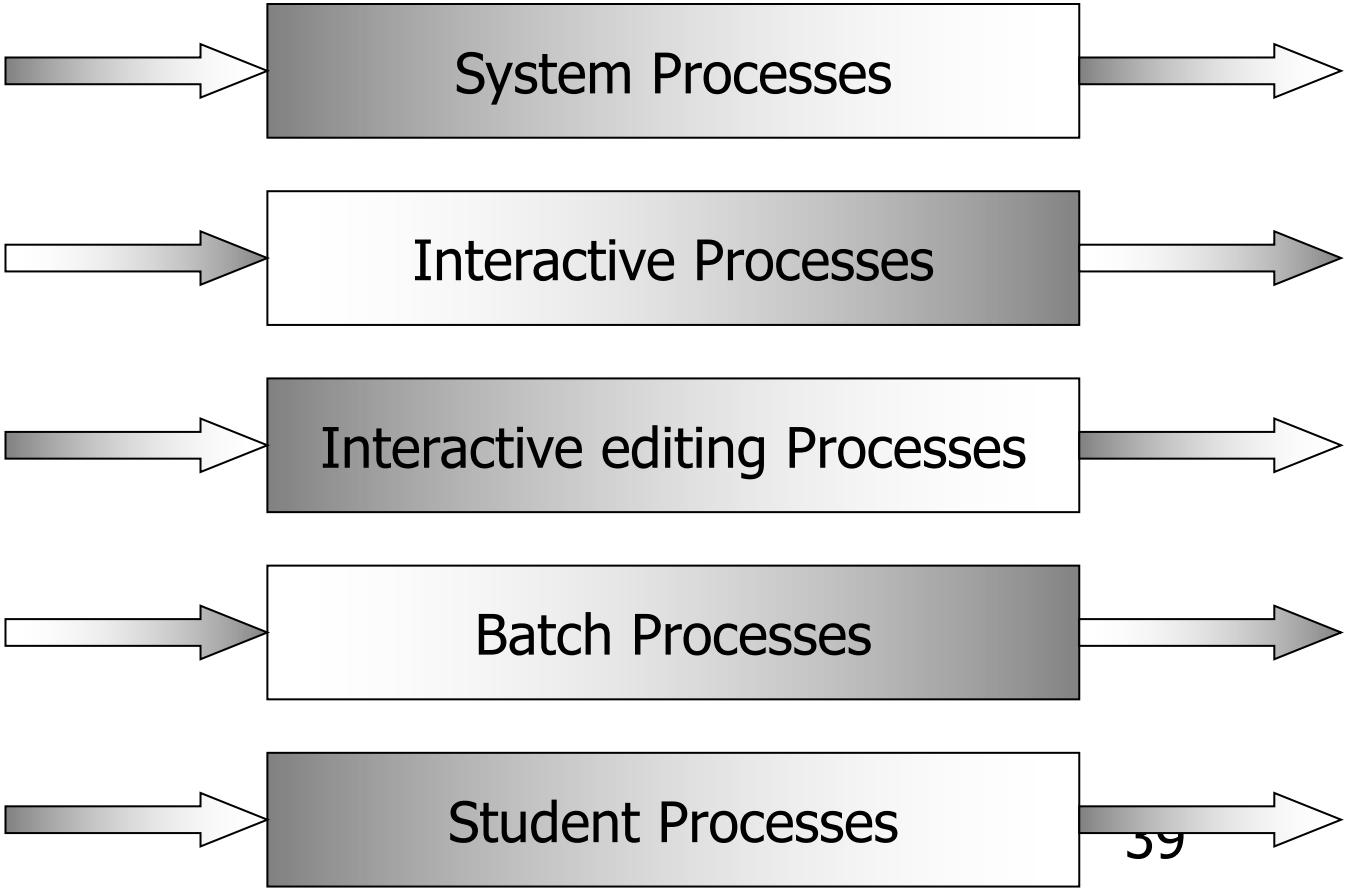
Time slice

- Each queue gets a fraction of CPU time to divide amongst its processes, e.g.,
  - 80% to foreground in RR
  - 20% to background in FCFS



Multilevel Queue Scheduling – Example

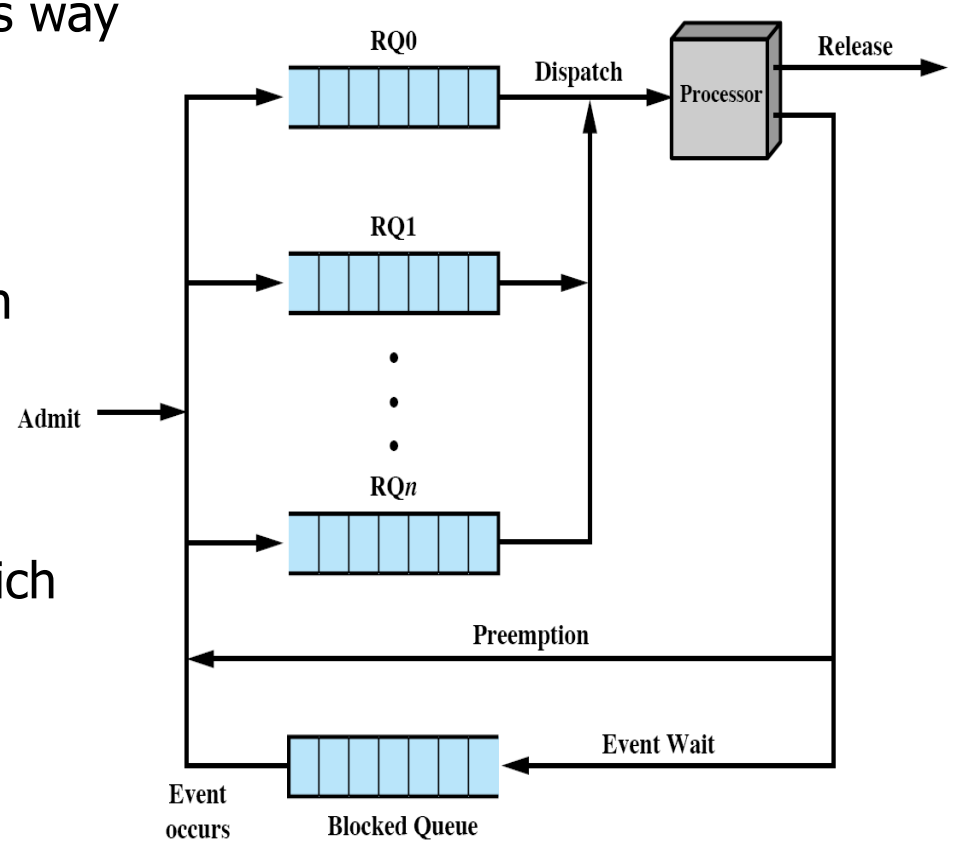
Highest Priority



Lowest Priority

Multilevel Feedback Queue

- A process can move between the various queues
  - Aging can be implemented this way
- Scheduling parameters
  - Number of queues
  - Scheduling algorithms for each queue
  - Method to upgrade a process
  - Method to demote a process
  - Method used to determine which queue a process enters first



Approximation of SJF: Multilevel Feedback Queue

Admission of P_i: $level(P_i) = 0$

- Assign process to RQ_0

Dispatch()

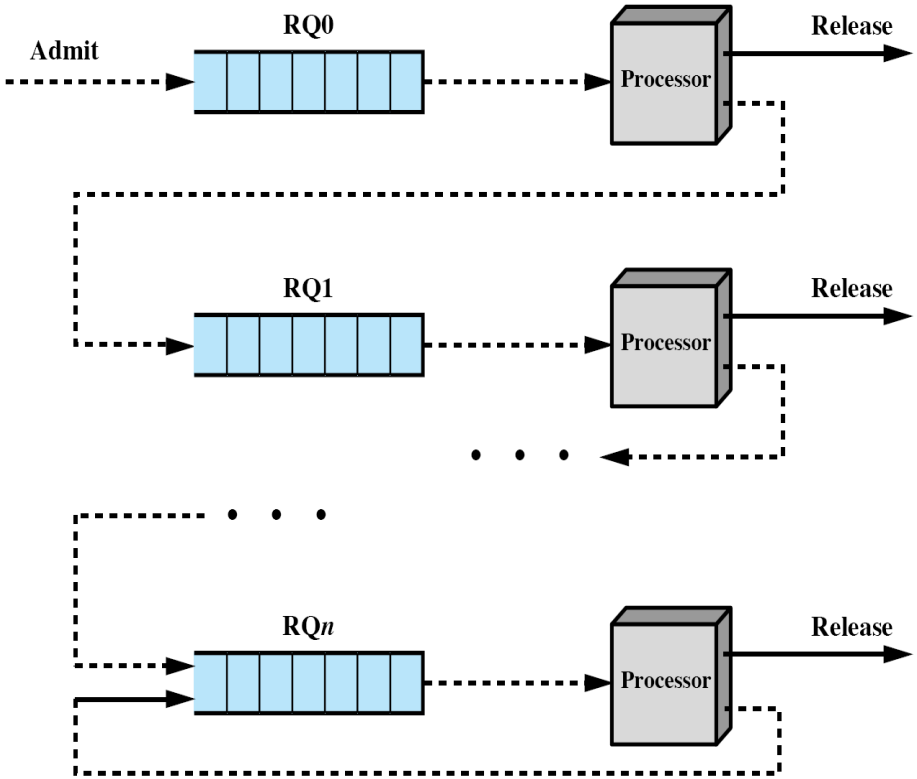
- Find smallest level s.t. $RQ_{level} \neq \emptyset$
- $P_j = RQlevel$.dequeue()
- Set time slice to fit level of process

Upon state change (P_i)

- Running → Ready
- $level(Pi) = \max\{level(Pi) + 1, n\}$
- RQ_{level}.enqueue(P_i)
- Dispatch()

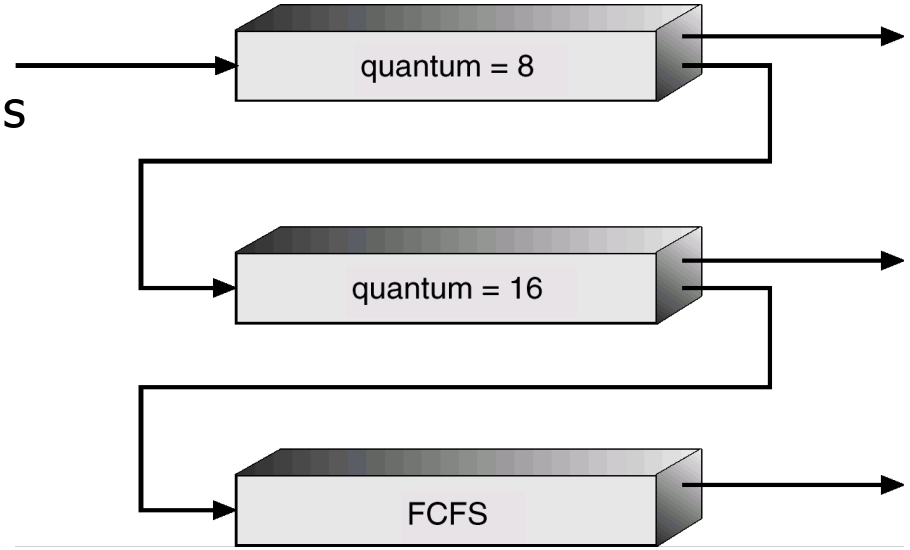
Upon state change (P_i)

- Running → blocked
- Enqueue P_i to Event Queue
- Dispatch()



Multilevel Feedback Queues – Example

- Three queues
  - Q_0 – RR with time quantum 8 ms
  - Q_1 – RR time quantum 16 ms
  - Q_2 – FCFS



Starvation is still possible here!

- Scheduling
  - A new process enters queue Q_0
  - When it gains CPU, process receives 8 ms
  - If it does not finish in 8 ms, process is moved to queue Q_1
  - At Q_1 process 16 additional ms
  - If it still does not complete, it is preempted and moved to queue Q_2

Summary Uniprocessor Scheduling

- Long, medium, short term schedulers
- Short-term scheduling criteria/optimization goals
  - CPU utilization, Throughput, Response time, Fairness, Scheduling overhead
- Short-term scheduling algorithms in uniprocessors
  - Basic algorithms: FCFS (Batch) vs. Round Robin (Interactive)
  - Shortest process next (optimal non-preemptive in avg. response time)
    - Determining length of next CPU burst (exponential averaging)
  - Dealing with starvation: Aging, e.g., HRRN
  - Efficient Scheduling: Multilevel Queues, e.g., approximate SPN

Multiprocessor Scheduling

Further reading:

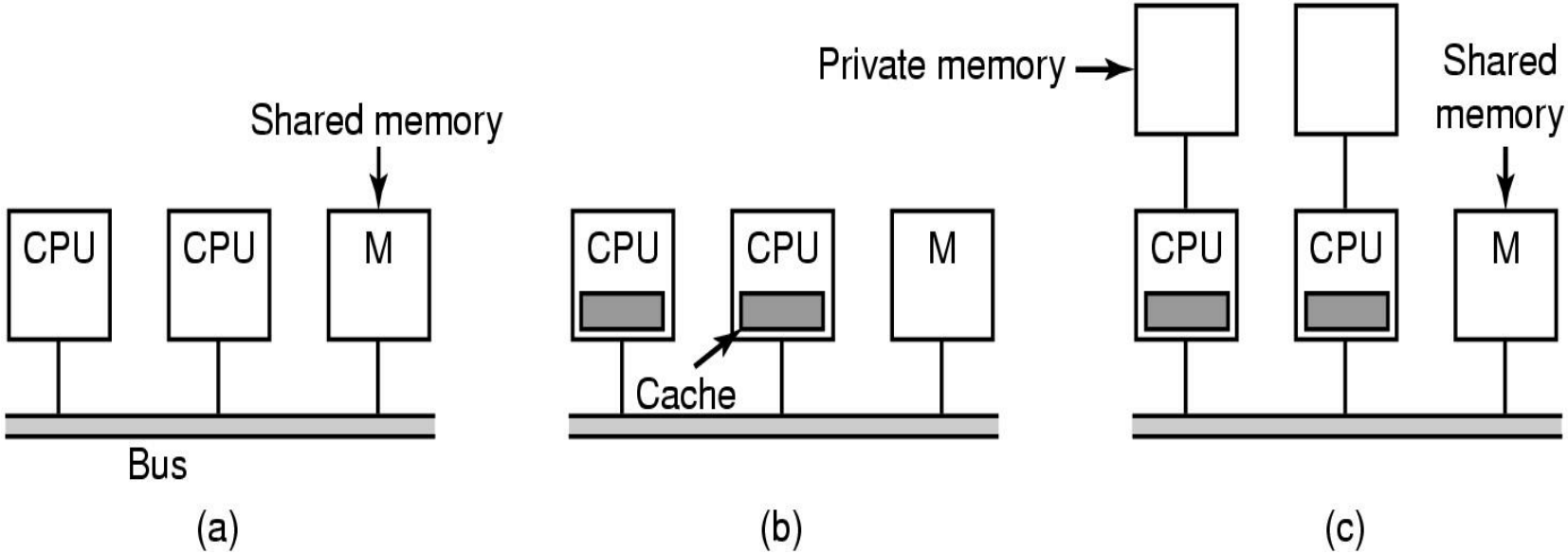
Chapter 10.1, Stallings

Chapter 5.4, Silberschatz

Chapter 8, Tanenbaum

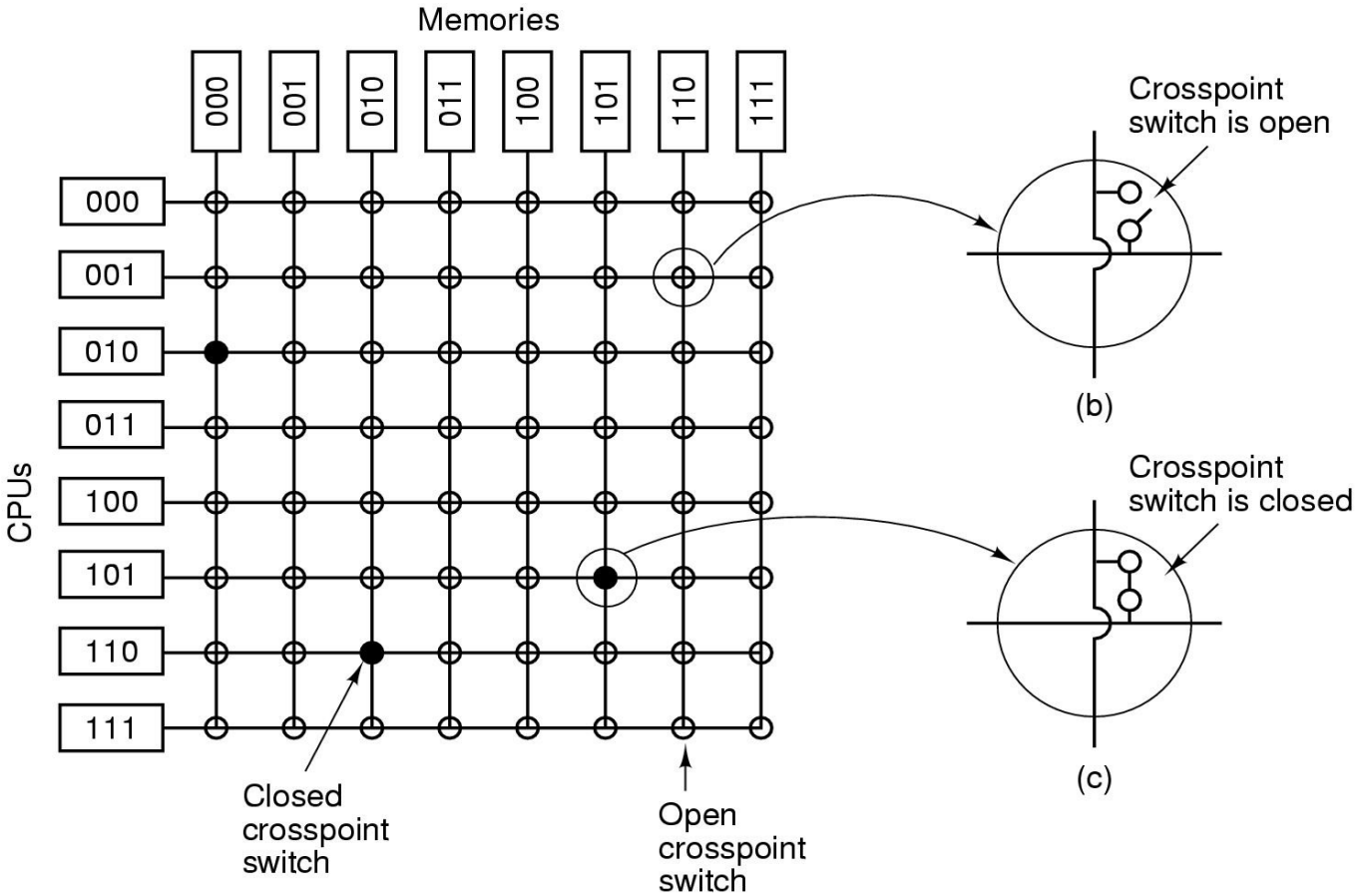
Multiprocessor System

A shared multiprocessor system allows the sharing of a common RAM by two or more CPUs



- Bus-based multiprocessors
  - All traffic goes via the common bus
  - Cache: can reduce bus access
  - Private memory: store program text, read-only data etc.

Multiprocessor Hardware



- Supports atomic read and write operations to memory
- Scalability issue: Number of crossbar switches increases in $O(m^2)$

Multiprocessor: Design Issues

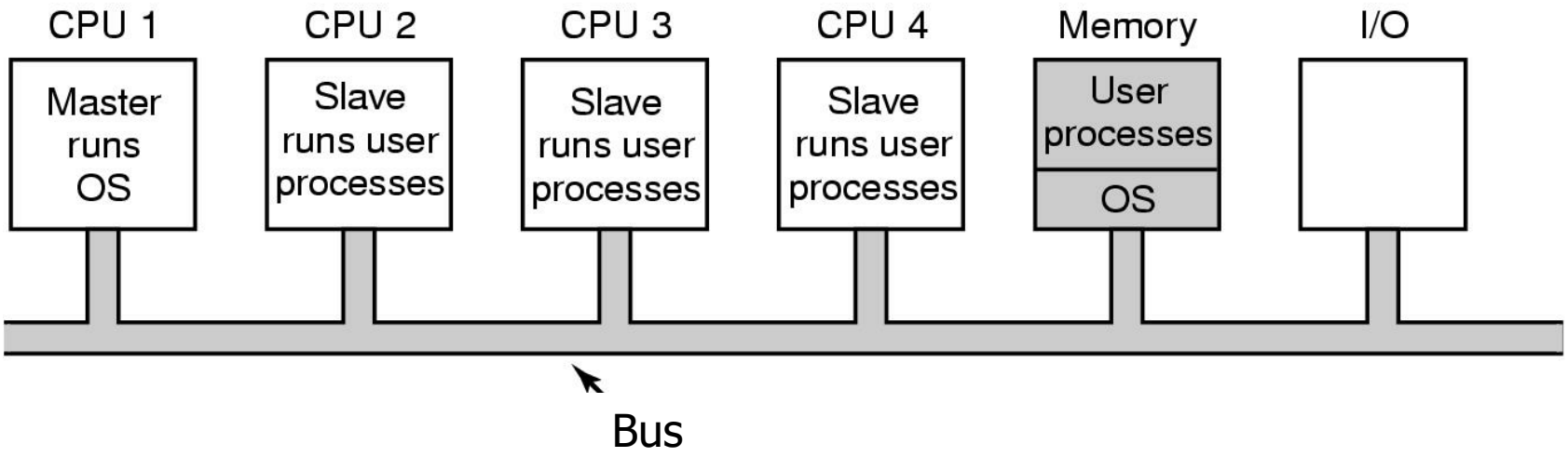
1. Execution of the OS scheduler
   - Master/Slave vs Peer Architecture
2. Assignment of processes to processors
   - Dedicated assignment of processes to processors
   - Processes migrate between processors
3. Multiprogramming

Design Issues: Where to Execute OS Functions?

- Master/Slave architecture: Key kernel functions always run on a particular processor
  - Master is responsible for scheduling
  - Slave sends service request to the master
  - Disadvantages
    - Failure of master brings down whole system
    - Master can become a performance bottleneck

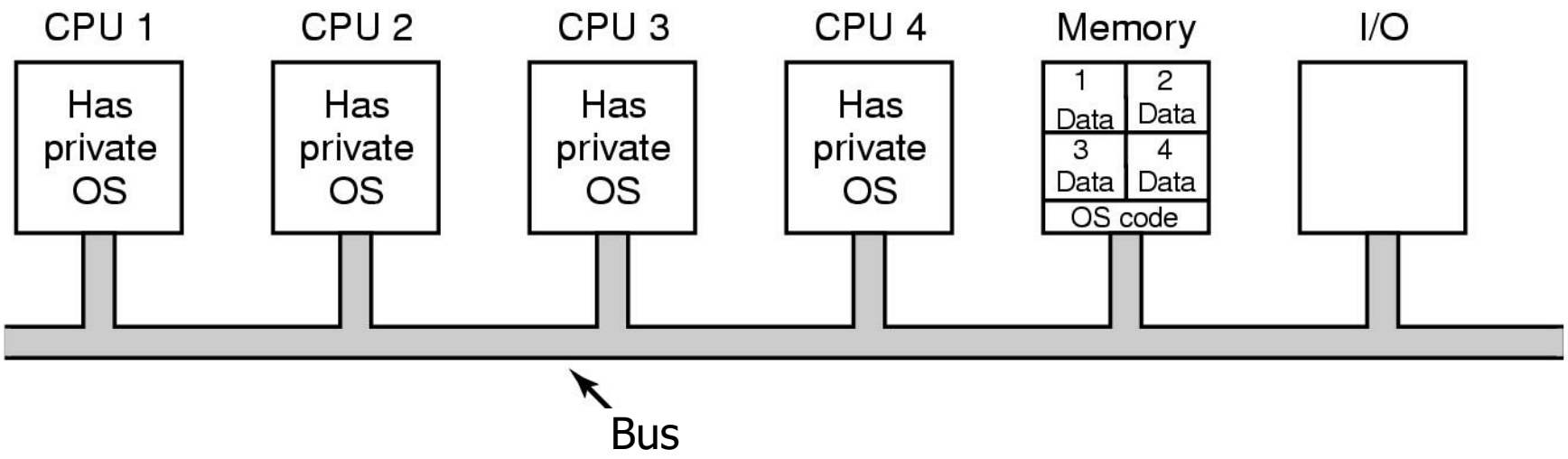
- Peer architecture: OS can execute on any processor
  - Each processor does self-scheduling
  - New issues for the operating system
    - Make sure two processors do not choose the same process

Master-Slave Multiprocessor OS



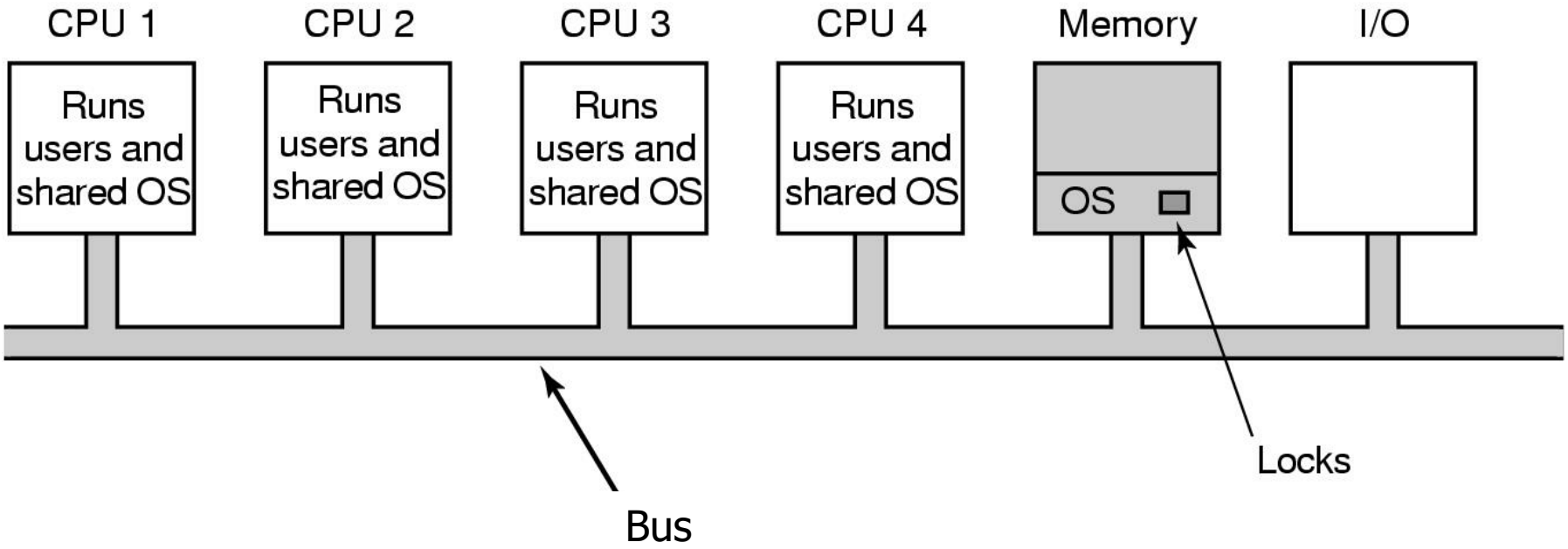
- Only one processor (master) hosts the operating system
- System calls are directed to the master
  - Many processors may be idle waiting for the master
  - Reduced CPU utilization

Non-symmetric Peer Multiprocessor OS



- Each CPU has its own operating system
- Feasible and simple approach
  - Processes on CPUs are independent
  - Cannot exploit unused resources of another processor

Symmetric Peer Multiprocessor OS



- Symmetric multiprocessors (SMP) model
  - Need to prevent possible conflicts (mutual exclusion)
  - Goal: have independent parts of the OS on different processors

Design Issues: Assignment of Processes to CPUs

Per-processor ready-queues vs. global ready-queue

- Permanently assign process to a processor
  - Less overhead
  - Processor affinity: processes execute faster when assigned to the same processor
    - First level cache preserves locality
  - A processor could be idle while another processor has a backlog

- Have a global ready queue and schedule to any available processor
  - Can become a bottleneck
  - Task migration not cheap
    - Loose locality of Cache

Design Issues: Multiprogramming

Recall: Tightly coupled multiprocessing (SMPs)

- Processors share main memory
- Controlled by operating system

Different degrees of parallelism

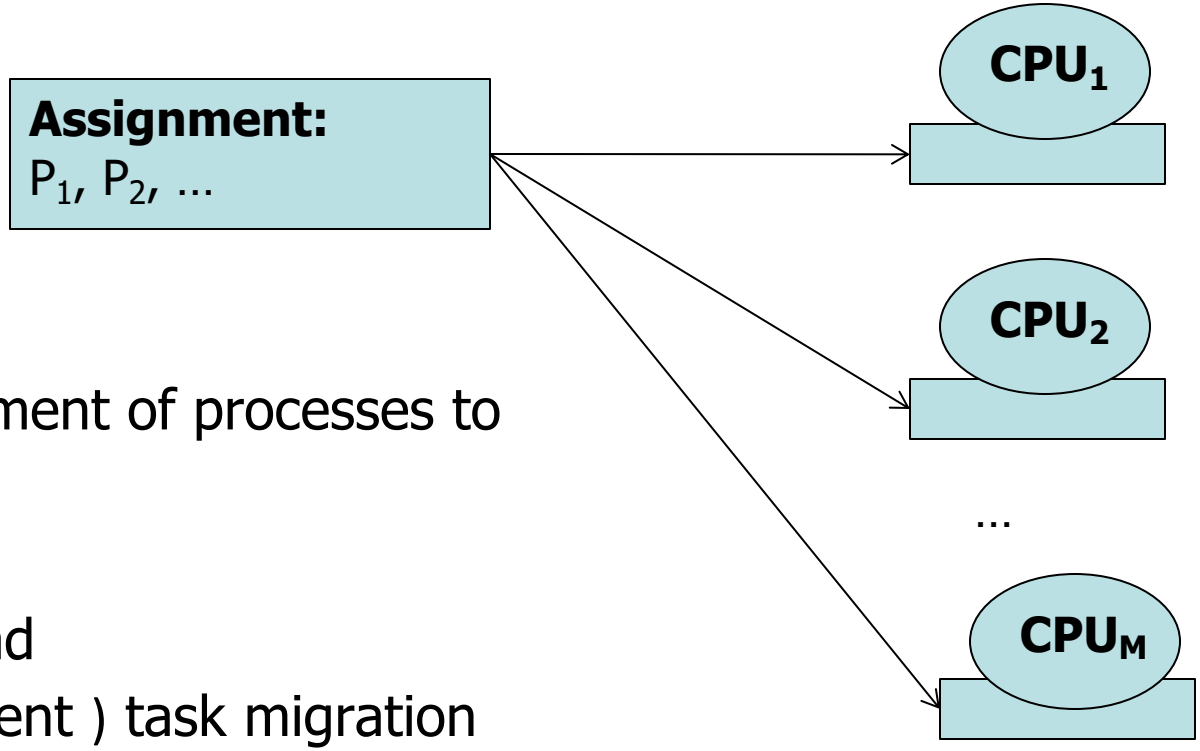
1. Independent and Coarse-Grained Parallelism
   - No or very limited synchronization
   - Can be supported on a multiprocessor with little change
2. Medium-Grained Parallelism
   - Collection of threads; usually interact frequently
3. Fine-Grained Parallelism
   - Highly parallel applications; specialized and fragmented area

Design issues: Multiprogramming

Experience shows:

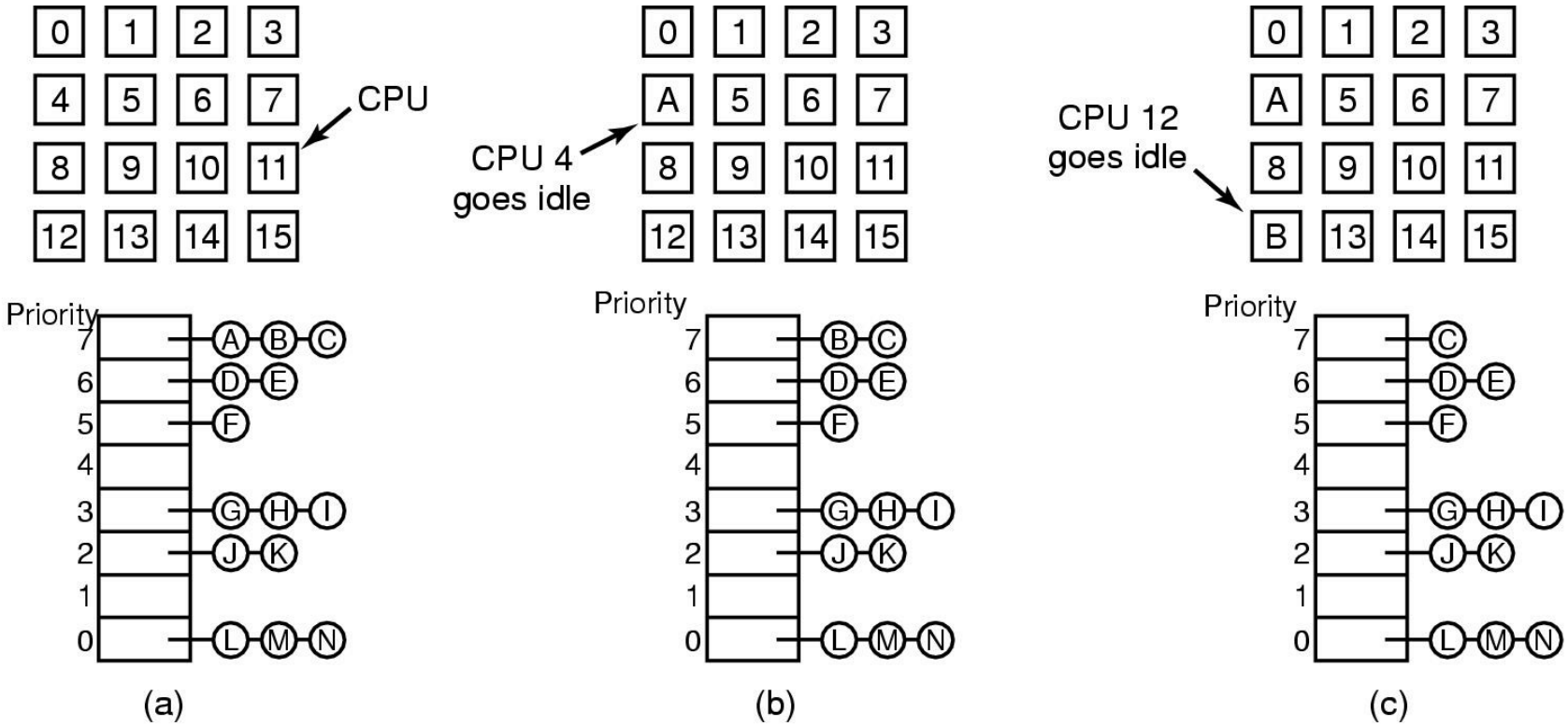
- Threads running on separate processors (to the extend of dedicating a processor to a thread) yields dramatic gains in performance
- Allocating processors to threads ~ allocating pages to processes
- Specific scheduling discipline is less important with more than one processors
  - the decision of "distributing" tasks is more important

Fixed assignment



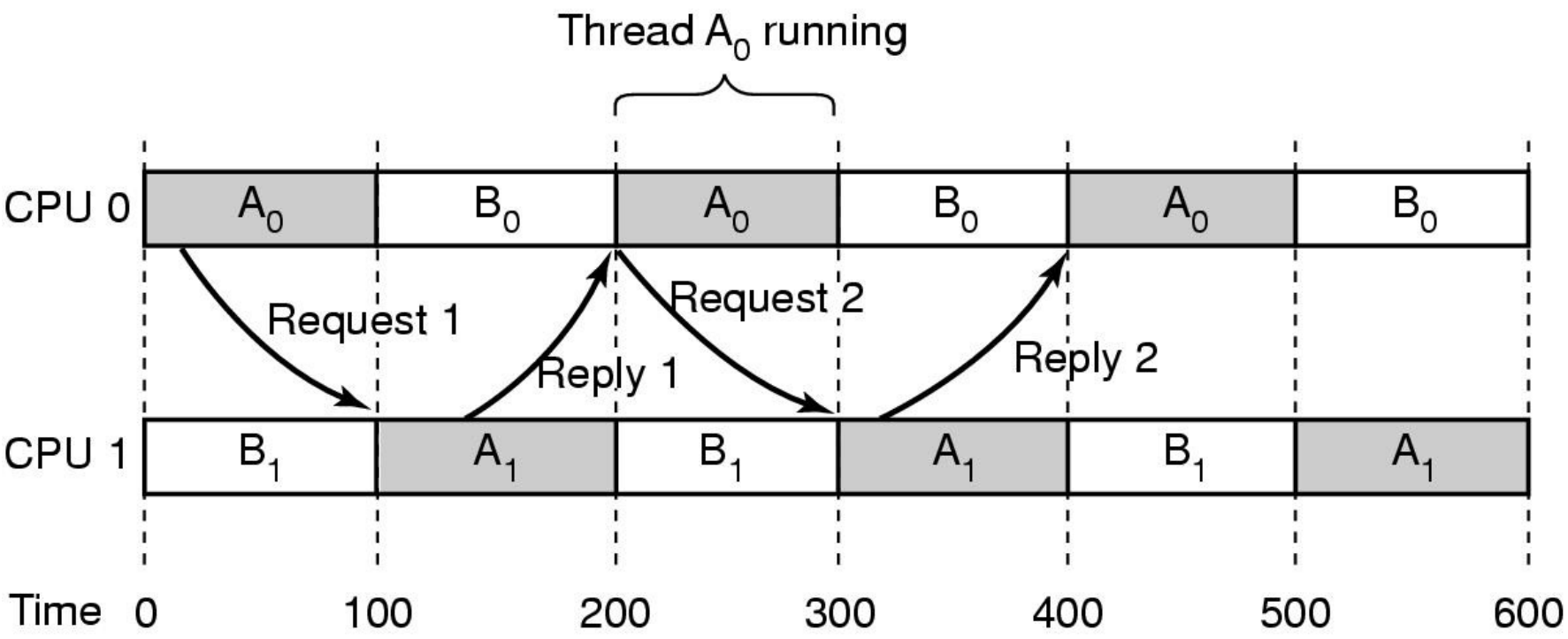
- Fixed assignment of processes to processors
- Efficient
- Less overhead
- Bad assignment) task migration

Multiprocessor Scheduling: Load sharing



- Single data structure for scheduling
- Policies: FCFS, prioritize job with smallest number of unscheduled threads, time sharing

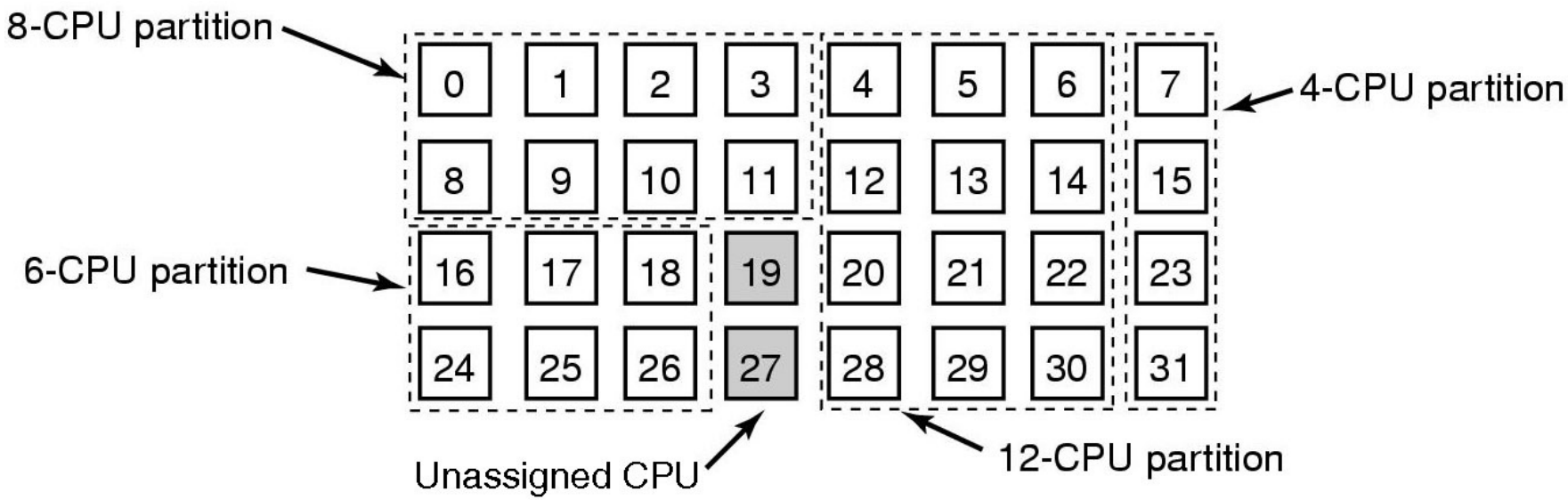
Problem with Load Sharing



- Interactions between two threads
  - Both belong to process A
  - Both running out of phase

Multiprocessor Scheduling: Space Sharing

- Form partitions of CPUs to run cooperating threads concurrently
  - All cooperating threads are deployed at the same time on the CPUs of a free partition
  - CPU occupied until all threads completed a CPU burst
- A group of threads may need to wait long until a sufficiently large partition becomes available
  - Limits the number of threads that can concurrently make progress



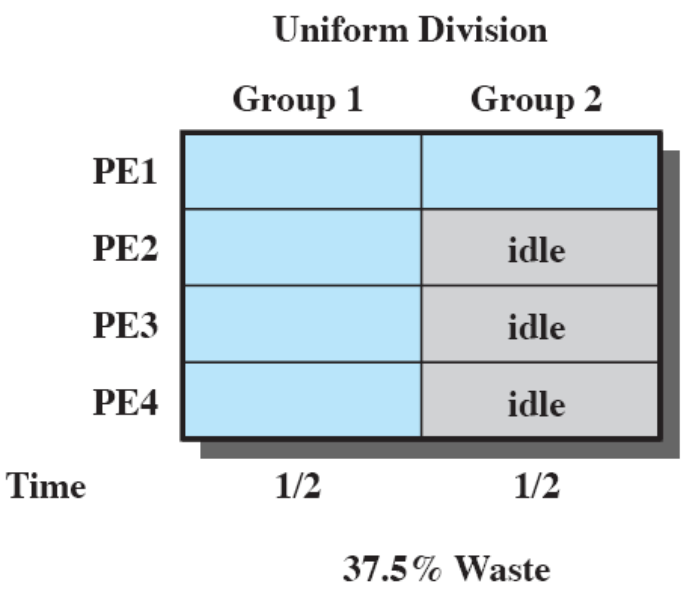
Gang Scheduling

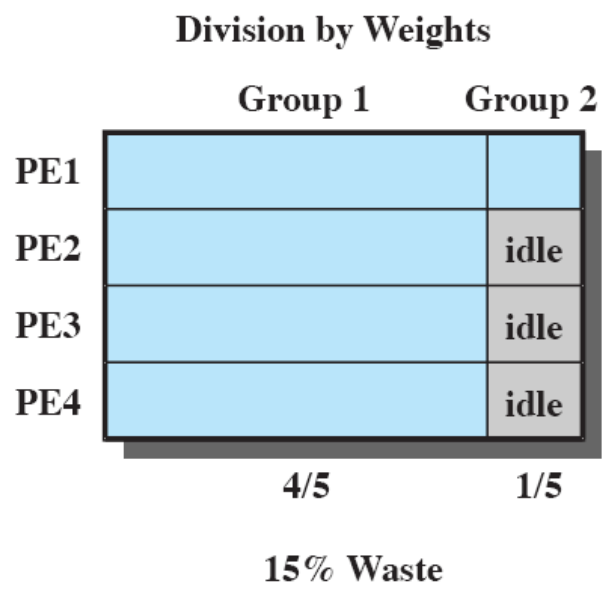
- Groups of related threads scheduled as a unit (a gang)
- All members of gang run simultaneously
  - On different timeshared CPUs
- All gang members start and end time slices together

Time slot	CPU 0	1	2	3	4	5
0	A_0	A_1	A_2	A_3	A_4	A_5
1	B_0	B_1	B_2	C_0	C_1	C_2
2	D_0	D_1	D_2	D_3	D_4	E_0
3	E_1	E_2	E_3	E_4	E_5	E_6
4	A_0	A_1	A_2	A_3	A_4	A_5
5	B_0	B_1	B_2	C_0	C_1	C_2
6	D_0	D_1	D_2	D_3	D_4	E_0
7	E_1	E_2	E_3	E_4	E_5	E_6

Gang Scheduling: Another Option

- Group threads
  - # threads per group < # processors
- Give each group a time slot
- Length of time slice chosen to minimize idle processor time





Summary: Multiprocessor Scheduling

Load sharing: processes/threads migrate dynamically

- Load is distributed evenly across the processors
- Needs central queue) performance bottleneck
- Bad use of processor affinity

Gang scheduling: Assigns threads to a particular processors

- Account for interactions between threads (e.g. synchronization)
- Extreme version: **Dedicated processor assignment** (no multiprogramming of processors)

Synchronization on shared data and competition on resources may rise additional problems : cf. Chapter Process Synchronization

Real-Time Scheduling

Further reading:

Chapter 10.2, Stallings

Chapter 5.5, Silberschatz

Chapter 7.4, Tanenbaum

Real-Time Systems

- Tasks or processes attempt to interact with outside-world events
  - Occur in "real-time"
  - Process must be able to keep up, e.g.,
    - Multimedia Applications
    - Control of laboratory experiments, Robotics, Air traffic control
    - Drive-by-wire systems, Tele/Data-communications,
    - Disaster command and control systems
- Correctness of the RT system depends **not** only on the logical result of the computation
  - **Time** at which the results are produced matters
  - Not necessarily high throughput) high performance computation
    - Tasks or processes come with a deadline (for starting or completion)
  - Requirements may be hard or soft

Hard vs. Soft Constraints

Hard constraints

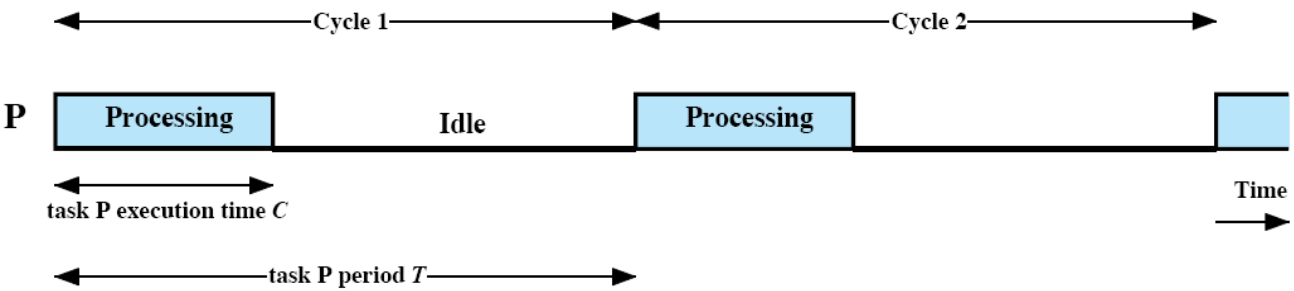
- If the system fails to fulfil a timing constraint) computational result is useless
  - Non critical: System functions with reduced performance
  - Critical: System cannot continue to function
  - Safety critical: Serious damage or loss of life
    e.g., breaks in a car, etc.
  - Verify correctness before starting a system!

Soft constraints

- Single time failures can be acceptable, however reduce the usefulness of a computation
  - Video streaming
  - Reservation systems
  - Statistical guarantees are often sufficient

Task Model

- Periodic Tasks
  - Task i arrives exactly at a given time interval T

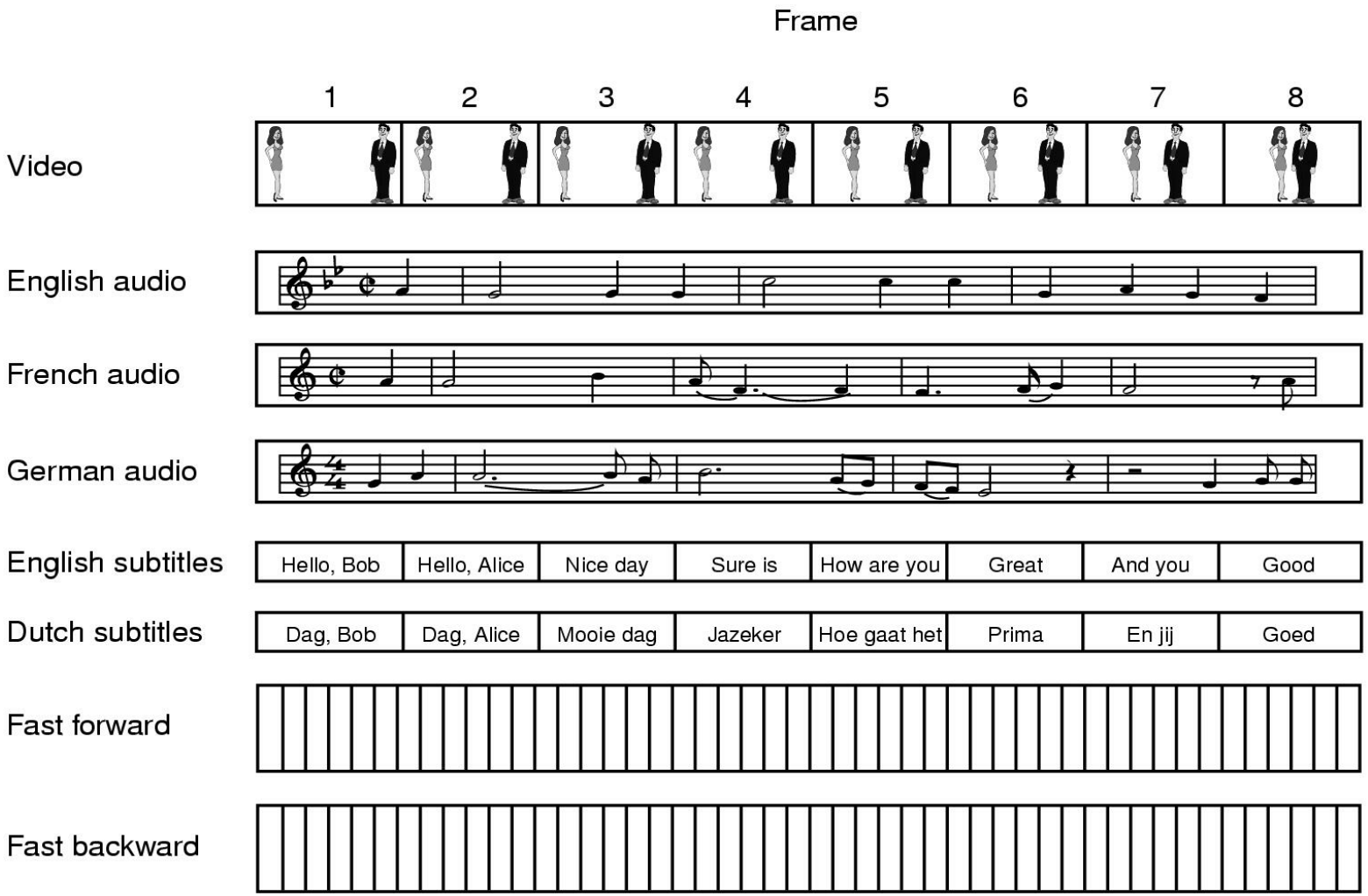


  - A task arrives with a time interval $\geq T$

- Aperiodic tasks
  - There is no minimum guaranteed time between two subsequent arrivals

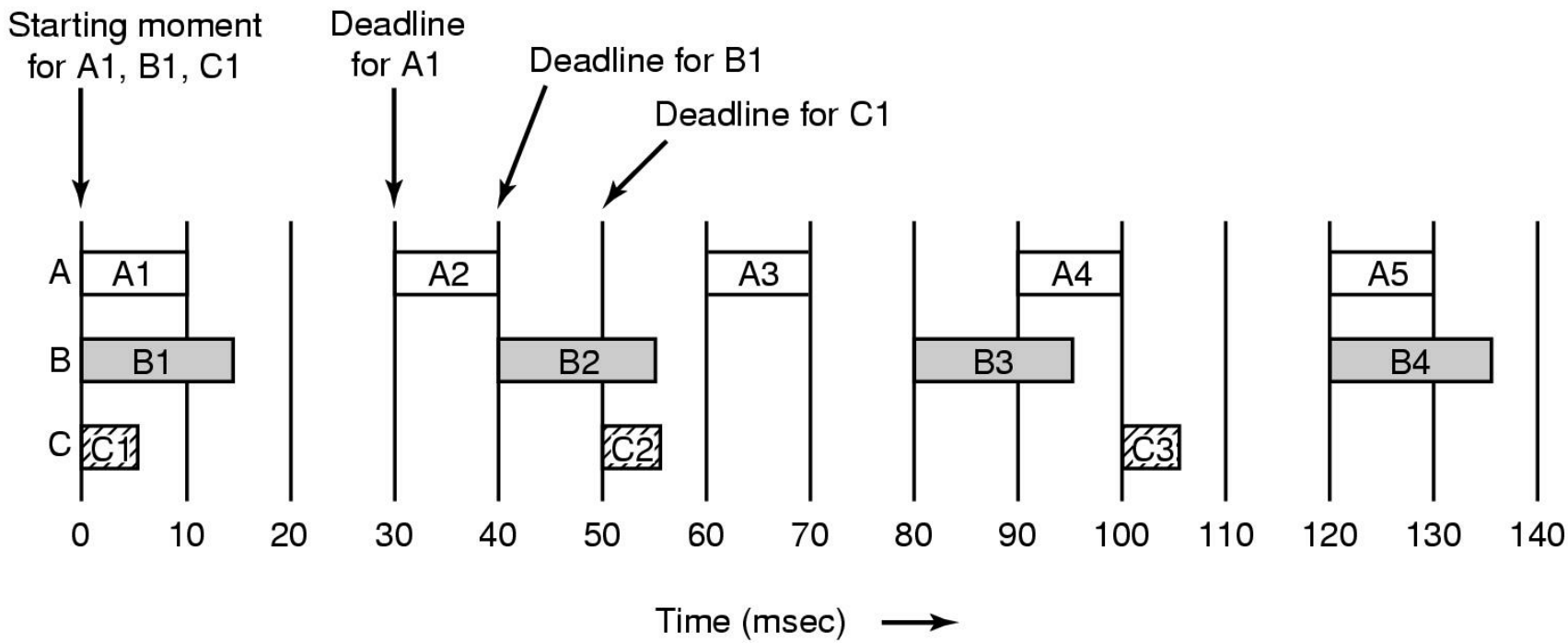
- Hard real time systems
  - Can consider only periodic and sporadic tasks

Example Multimedia Process Scheduling



A movie may consist of several files

Example Multimedia Process Scheduling



- Periodic processes displaying a movie
- Frame rates and processing requirements may be different for each movie (or other process that requires time guarantees)

Scheduling in Real-Time Systems

Schedulable real-time system (one processor)

- Given
  - *n* periodic tasks
  - Task *i* occurs within period T_i and requires C_i seconds
- Then the load can only be handled if

$$\text{Utilization} = \sum_{i=1}^{n} \frac{C_i}{T_i} \leq 1$$

EDF and RM Scheduling

Earliest Deadline First (EDF)

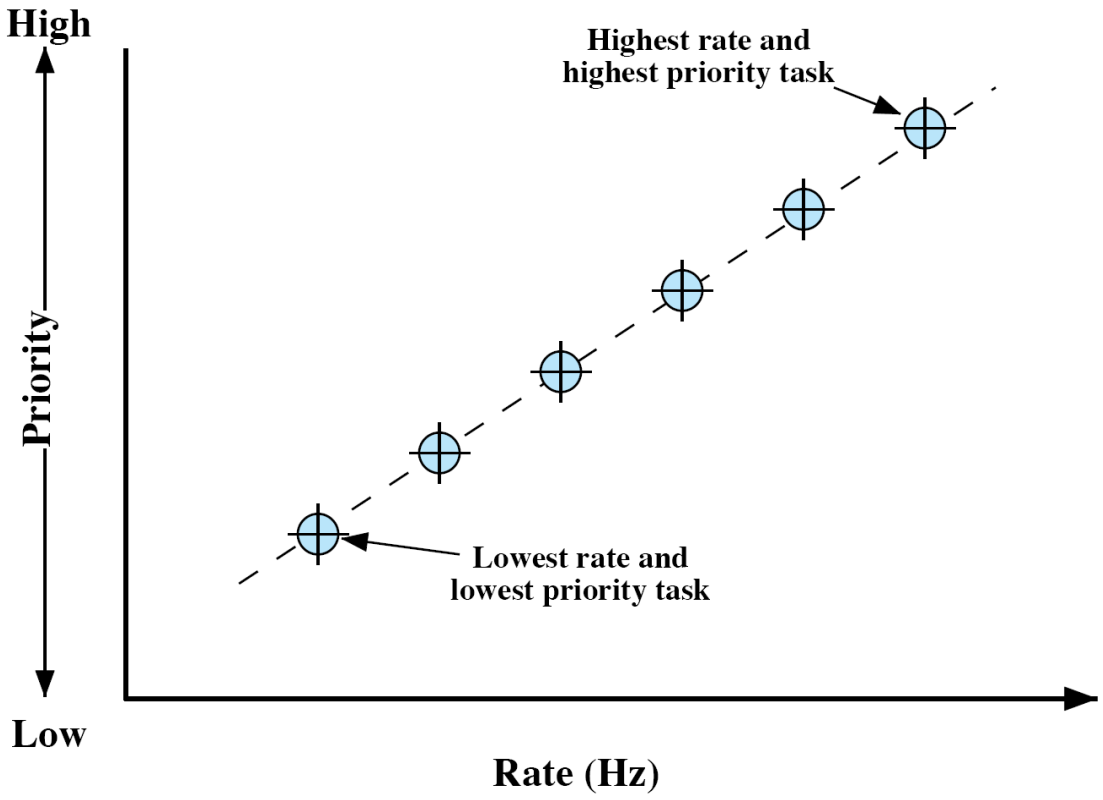
- Schedule periodic task with earliest deadline first
- Optimal for dynamic assignment of priorities (uniprocessor)
- Highest priority defined by deadline

Rate Monotonic Scheduling (RM)

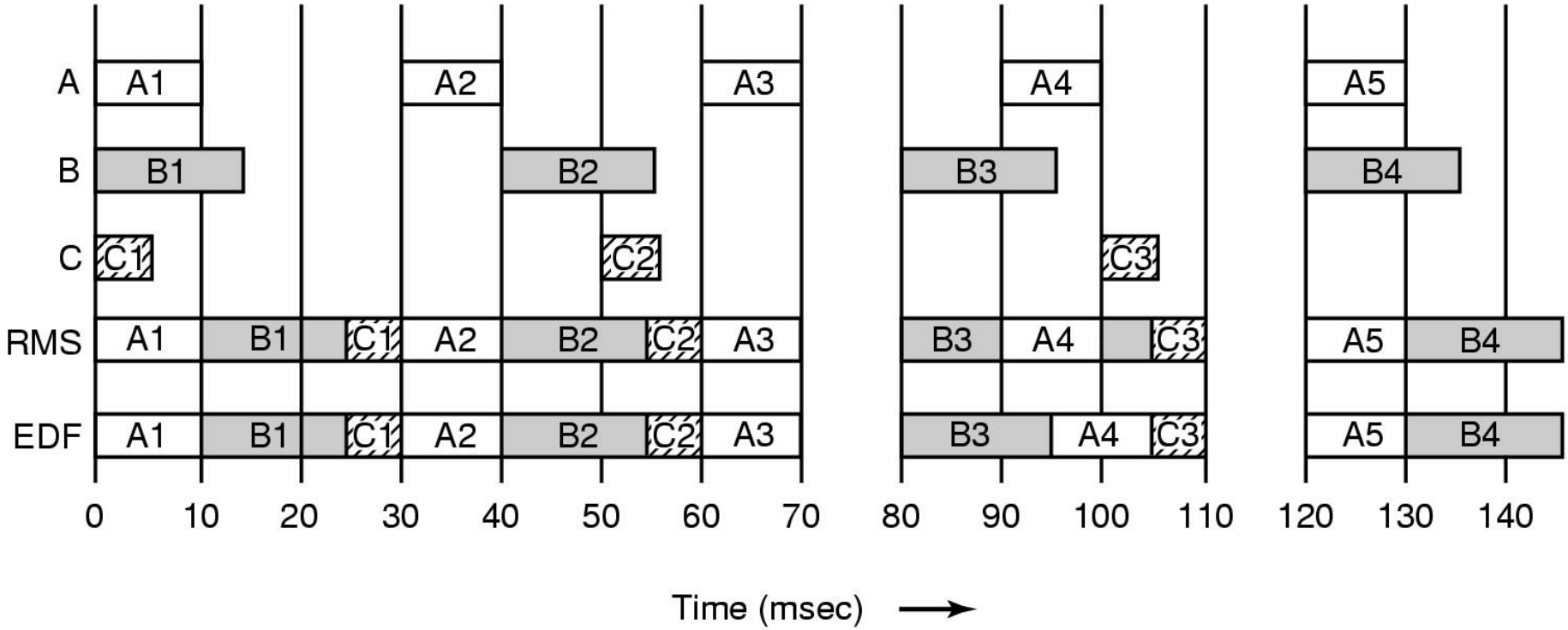
- Schedules tasks with shortest period first
- Optimal for a static assignment of priorities (uniprocessor)
  - These are the tasks with highest priority

Rate Monotonic Scheduling

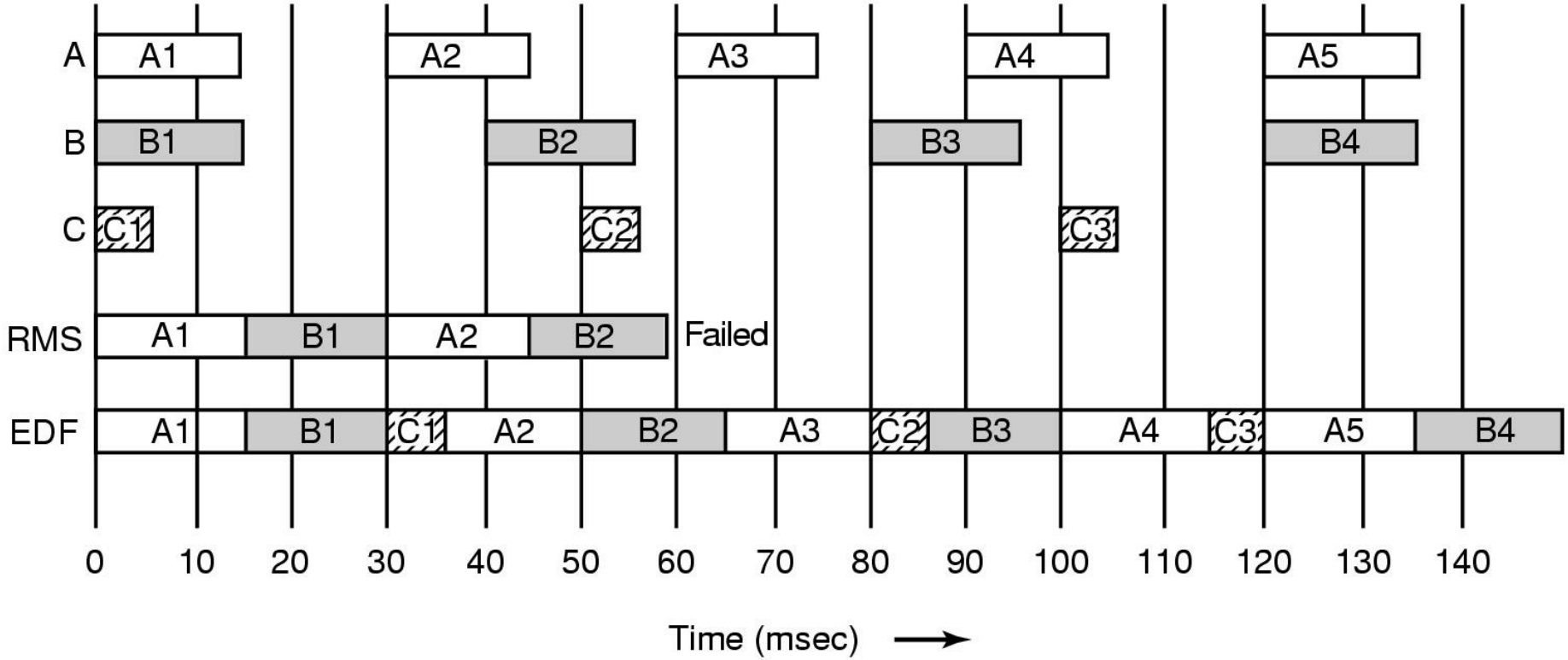
- Assigns priorities to tasks on the basis of their periods
- Highest-priority task is the one with the shortest period



Example Using EDF and RMS



Example Using EDF and RMS



Another example of real-time scheduling with RMS and EDF

Can We Determine When a Task Set is Feasible?

- Deciding the feasibility of a task set is NP complete for many instances
  - Offsets and deadlines
  - Minimize late tasks

- Periodic task sets can use process utilization
  - RMS succeeds for a periodic task set iff

$$\sum_{i=1}^{n} \frac{C_i}{T_i} \le n(2^{1/n} - 1)$$

  - A task set is schedulable iff it is also EDF schedulable

$$\sum_{i=1}^{n} \frac{C_i}{T_i} \le 1$$

n	$n(2^{1/n} - 1)$
1	1.0
2	0.828
3	0.779
4	0.756
5	0.743
6	0.734
•	•
•	•
•	•
∞	$\ln 2 \approx 0.693$

EDF or RMS?

- RMS "accommodates" task set with less utilization
  - Recall: for EDF that is up to 1
- RMS is often used in practice
  - Stability is easier to meet with RMS when a system cannot meet all deadlines
  - Priorities are static, under transient period with deadline-misses
    - Critical tasks can be "saved" by being assigned higher (static) priorities
  - It is suitable for combinations of hard and soft RT tasks

Any Question So Far?